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Test beam analysis of LGADs for the High Granularity Timing Detector

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Abstract

In the upcoming High-Luminosity phase of the LHC, the ATLAS experiment will be upgraded with a new detector system: the High Granularity Timing Detector. The HGTD will be the first large-scale detector to provide timing information to ATLAS, deploying a new technology: Low Gain Avalanche Detectors (LGAD), thin silicon diodes with a mild internal charge multiplication. According to the HGTD's designed performance, LGADs are required to have a collected charge of at least 4 fC, an efficiency of 95%, and a time resolution of 35 ps (70 ps) at the beginning (end) of their life.

This thesis presents an analysis of the May 2023 test beam campaign, conducted at CERN, using a 120 GeV pions beam provided by SPS. Devices from three different vendors were tested: IHEP-IME, USTC and CNM. The devices were tested at -30°C , the temperature of the HGTD in operation. The sensors were irradiated to different fluences, up to $2.5 \times 10^{15} \text{ n}_{\text{eq}}$, corresponding to the maximum expected dose for a single sensor during its time in operation.

All sensors irradiated up to $1.5 \times 10^{15} \text{ n}_{\text{eq}}$ satisfied the HGTD's requirements at bias voltages lower than 400 V. At the highest fluence, the LGADs did not meet the charge and efficiency requirements, but still achieved a time resolution below 70 ps. Additionally, the sensors were tested at angles of 0° to 14° to the beam line, which showed a small but consistent improvement in charge collection and time resolution.

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Introduction

In the coming years, the LHC will undergo a series of upgrades to accommodate the requirements of the High-Luminosity phase [1], aimed at enabling more precise measurements and improving sensitivity to rare processes beyond the Standard Model.

In particular, the ATLAS experiment will have many of its subdetectors improved and/or replaced. Additionally, to help deal with the problems caused by the increased luminosity, a new detector system will be integrated into the experiment: the High Granularity Timing Detector [2]. The HGTD will provide time information of charged particles, helping differentiate between collisions that happen very close in space but well separated in time. To achieve this, a special type of silicon sensors will be employed: Low Gain Avalanche Detectors [3]. These sensors combine very fast response with an internal gain, to enhance the signal and better endure the high radiation environment that they will face.

To achieve the HGTD's precision targets, the LGADs need to satisfy some requirements. The sensors must collect enough charge for the signal to be measured by the electronics ($\approx 4\text{fC}$), they must have a time resolution better than 35 ps at start of life (and up to 70 ps at end of life), and, finally, they must have an efficiency per hit greater than 95%.

There exist several vendors and version of LGADs so, to test and compare their performances, the sensors are typically put through a collimated beam of accelerated particles, simulating the environment of the ATLAS detector. In this thesis we focused on the test beam campaign of May 2023, conducted at one of CERN's facilities, where LGADs from three different vendors were characterized: USTC (University of Science and Technology of China), IHEP (Institute of High Energy Physics, China) and CNM (Centro Nacional de Microelectrónica, Spain).

In the test beam a EUDET-type beam telescope of 6 silicon pixel detectors was used for track reconstruction, a cooling box maintained the LGADs at -30°C and another device had the role of reference for the time resolution measurements.

After some pre-processing, the data was further refined to suppress various sources of noise. For each sensor, the time resolution, the collected charge and the efficiency were calculated. The sensors were grouped based on version, vendor and level of radiation for easier comparison. Additionally, some interesting effects that emerged during the analysis were further investigated.

The first chapter of this thesis describes in general the LHC and the ATLAS experiment, the next chapter focuses on the HGTD and the silicon sensors it will use: the LGAD. The following chapter describes the key elements in the test beam setup used for the data taking. The fourth chapter follows the implementation of the analysis, providing the reasons and the tools used. Finally, the fifth chapter presents the general results of the analysis. Some additional information, plots and tables are provided in the Appendix.

1 | The LHC and the ATLAS experiment

The theory that currently best describes the fundamental particles and their interactions is known as the Standard Model [4]. It was developed during the 20th century and finalized in the mid-1970s, after experimental confirmations of quarks. The Standard Model has since predicted and explained a wide range of phenomena with great accuracy, important examples are the W and Z bosons and the Higgs boson. However, some questions still remain unanswered.

1.1 The Large Hadron Collider

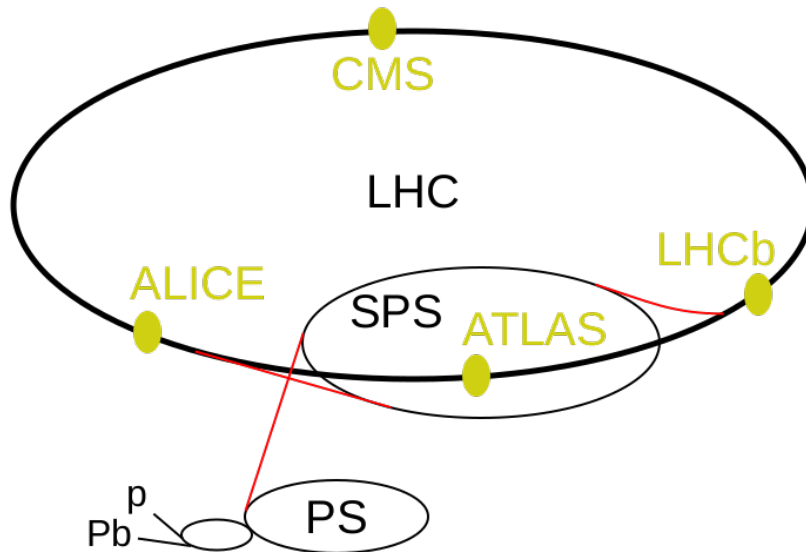


Figure 1.1: Simplified scheme of the CERN accelerator complex. Each proton or lead ion starts from a linear accelerator (LINAC 4 or LINAC 3, respectively, not in the picture), it is then transferred to the Proton Synchrotron (SP), into the Super Proton Synchrotron (SPS) and finally into the Large Hadron Collider (LHC). During each phase of acceleration the particles reach higher momentum before being transferred to the next stage.

The Large Hadron Collider (LHC) is the largest particle accelerator in the world. It was built between 1998 and 2008 by the European Organization for Nuclear Research (CERN, *Conseil européen pour la Recherche nucléaire*) in collaboration with more than 100 countries with the purpose of colliding hadrons¹ at high

¹Hadrons are particles that are made up of two or more quarks, e.g. protons and neutrons

energies and study the products of these collisions. It consists of a 27 km ring at roughly 100 m of depth situated at the Franco-Swiss border, near Genève. Two beams of particles travel at near light speed in opposite directions inside two pipes surrounded by superconducting magnets. At specific locations the beams converge and collisions take place, these events can then be observed with specialized detectors built along the ring.

1.1.1 The research goals

The construction of the LHC was motivated by the need for experimental data to study the laws of our universe. In particular, its main objectives were [5]:

- The search for the Higgs boson: the particle (whose mass could not be predicted) responsible for the origin of mass, culminated in the discovery of the Higgs boson in 2012 [6].
- Supersymmetry: a theory that hypothesizes the existence of massive partners of the particles in the current Standard Model.
- Dark matter and dark energy: the matter that we know only makes up a small fraction, $\approx 4\%$, of the content of the universe, and the particles or phenomena responsible for the remaining 23% (dark matter) and 73% (dark energy) are still unknown.
- Matter-antimatter asymmetry: matter and antimatter should have been produced in equal amounts during the Big Bang, yet observations show an overwhelming prevalence of normal matter.
- Quark-gluon plasma: a state of matter that existed in the early stages of the Universe, before ordinary matter could emerge.

The LHC is part of the CERN accelerator complex, a chain of machines in which each link accelerates a beam of particles (protons or heavy ions) to higher energies and feeds it to the next one. A simplified version of the complex is shown in Figure 1.1. The LHC began operations in 2010 with energies of 3.5 TeV, increased up to 4 TeV in 2012. After two years of Long Shutdown for upgrades and maintenance, activities resumed in 2015 with Run 2, and beam energies of 6.5 TeV ($\sqrt{s} = 13$ TeV, center of mass energy) were achieved.

1.1.2 The main experiments

Along the accelerator tracks there are four main experiments (Figure 1.1):

- ALICE (A Large Ion Collider Experiment): dedicated to heavy-ion physics [7].
- ATLAS (A Toroidal LHC Apparatus): one of the two general-purpose detectors, able to investigate a wide range of physics [8].
- CMS (Compact Muon Solenoid): the second general-purpose detector of LHC [9].
- LHCb (Large Hadron Collider beauty): an asymmetric detector that observes mainly forward particles to study the "b" quark [10].

Additionally, there are several other experiments part of the CERN complex, including test beam facilities in the North Area which use particle beams from SPS to perform a variety of tests.

One of the most important parameters of a collider, along with the beam energy, is the luminosity. It is used to quantify the amount of interactions that will occur and, therefore, the expected rate of events produced by the collisions. For two bunches² colliding head-on with frequency f_{coll} , the luminosity can be calculated by:

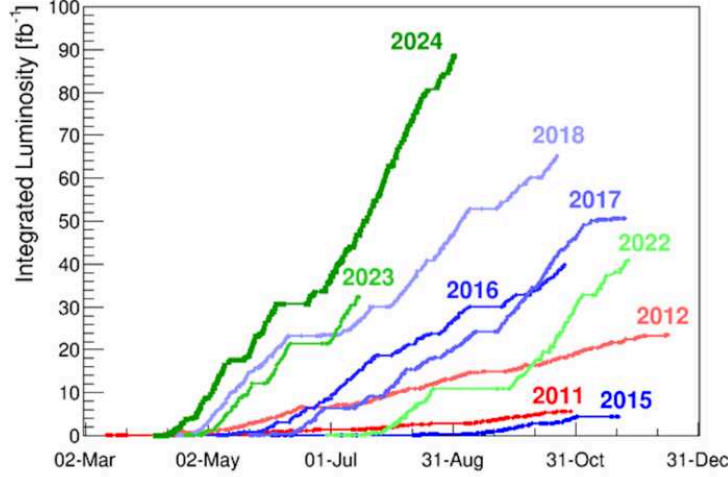


Figure 1.2: Overview of the integrated luminosity as a function of the date for each year of LHC running, with 2024 greatly exceeding all other years. From [11].

$$\mathcal{L} = f_{coll} \frac{N_1 N_2}{4\pi\sigma_x\sigma_y} \mathcal{F} \text{ [cm}^{-2}\text{s}^{-1}\text{]}. \quad (1.1)$$

Where N_1 and N_2 are the number of particles in each bunch, σ_x and σ_y define the transverse spread of the bunches (approximately normal distributions) at the interaction point, and $\mathcal{F}$ is a factor of order 1 to account for inefficient geometric overlapping and other effects.

A related quantity is the integrated luminosity, which is the integral over time of the instantaneous luminosity. It quantifies the amount of data that is available to be studied, and it is typically measured in inverse femtobarns $\text{fb}^{-1} = 10^{-28} \text{ m}^{-2}$.

After the second Long Shutdown that took place between 2018 and 2022 LHC has been operating (Run 3) and already delivering record-breaking integrated luminosity (Figure 1.2).

1.1.3 The High-Luminosity upgrade

In the upcoming years there will be a third Long Shutdown (from around 2026, until early 2029) to prepare for the next phase: the High-Luminosity LHC Project (HL-LHC) [1]. The plan aims to increase the instantaneous luminosity up to $1.5 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ (compared to $2.1 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ of Run 2 [12]) and to bring the integrated luminosity up to 3000 fb^{-1} in the following 10-12 years. This will enable more accurate measurements of new particles and the possibility to observe rare processes below the current sensitivity level. On the other hand, this upgrade will introduce a wide range of new challenges for all the systems within CERN's accelerator complex. For instance, the detectors will be exposed to higher radiation doses, which may impact performance, and the increased number of collisions will lead to more pronounced pile-up effects complicating track reconstruction.

1.2 The ATLAS experiment

ATLAS (**A** Toroidal LHC Apparatu**S**) is a general purpose detector built for probing p - p and A - A collisions (*proton-proton* and *heavy ion-heavy ion*, respectively). Currently, the experiment's main detector systems are: the Inner Detector, the Calorimeters, the Muon Spectrometer and the Magnet System [14].

²The particles in the LHC beam do not travel uniformly distributed, rather they are grouped in many *bunches*, synchronized to intersect at the center of each main detector.

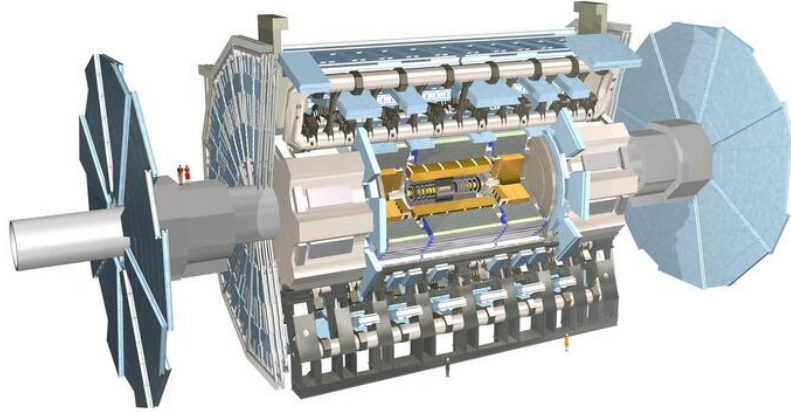
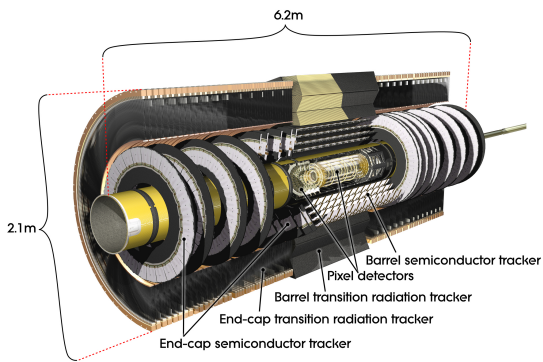
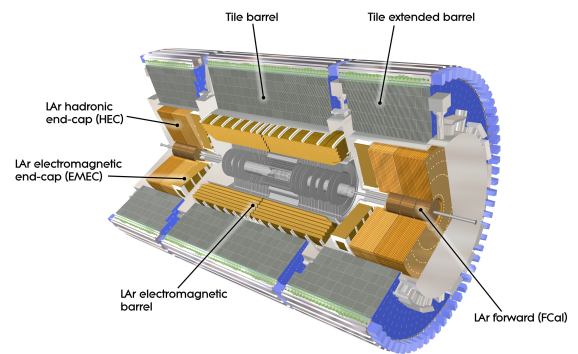


Figure 1.3: Overview of the ATLAS detector. From [13]



(a) The Inner Detector and its three components: the Pixel Detector, the Semiconductor Tracker and the Transition Radiation Tracker. From [15].



(b) The Calorimeters: the Liquid Argon Calorimeter (LAr), comprising the barrel, electromagnetic and hadronic end-cap, and forward sections, and the Tile Hadronic Calorimeter, divided into barrel and extended barrel sections. From [16].

Figure 1.4: Cutaway views of the ATLAS Inner Detector (left) and Calorimeter system (right).

1.2.1 Inner Detector

The main purpose of the Inner Detector is to reconstruct the vertices of the collisions, it comprises three sub-detectors (Figure 1.4a).

The Pixel Detector is the first point of detection, only 3 cm away from the LHC beam line, and it is made of four layers of silicon pixels, more than 92 million in total. Charged particles originating from the collision points leave a small amount of energy that can be measured by the pixels with a precision of almost 10 μm . These signals are then used to reconstruct the origin and the momentum of the particles.

The Semiconductor Tracker surrounds the Pixel Detector and complements it in the detection and reconstruction of the tracks of charged particles. It consists of over 6 million "micro-strips" silicon detectors with a precision of up to 25 μm .

The final layer of the Inner Detector is the Transition Radiation Tracker (TRT). Unlike its neighbouring silicon sub-detectors, TRT is composed of 300 000 4 mm diameter tubes with a 30 μm gold-plated tungsten wire in the center and filled with a mixture of gas. As charged particles pass through the volume they ionize the gas, which creates detectable electric signals. The location can then be determined, helping with the tracks reconstruction and the momentum measurement. Additionally, the moving particles emit transition radiation³, and the intensity of this radiation depends on the γ Lorentz factor, thus allowing to distinguish between heavier and lighter particles.

1.2.2 Calorimeter

Calorimeters are designed to absorb most of the incoming particles, forcing them to deposit all of their energy and stop within the detector. They are divided into two categories: electromagnetic and hadronic calorimeters. Electromagnetic calorimeters (ECAL) measure the energy of electrons and photons while they interact with matter via electromagnetic force. Hadronic calorimeters, on the other hand, measure the energy of hadrons¹ as they interact with atomic nuclei via strong force. Calorimeters can stop most known particle, except for muons⁴ (observed with the Muon Spectrometer) and neutrinos.

The ATLAS calorimetry system uses two different technologies: the Liquid Argon Calorimeter (LAr) and the Tile Hadronic Calorimeter.

The Liquid Argon Calorimeter (LAr) features layers of metal (either tungsten, copper or lead), with liquid argon in-between, kept at a temperature of -184°C . Several modules are placed cylindrically around the beam (the barrel) and at the extremities (endcaps). The LAr forward (FCal) covers the region close to the beam using three disks of metal (copper, tungsten and tungsten).

The Tile Hadronic Calorimeter surrounds the LAr calorimeter, it has the goal of measuring hadronic particles that did not deposit all of their energy in the LAr. It consists of alternating layers of steel and plastic scintillating tiles. Particles hitting the steel layer create a shower of other particles, and the plastic scintillators in turn produce photons, which are then converted into an electric current to determine the original particle's energy.

1.2.3 Muon Spectrometer

The outer layer of the ATLAS experiment is composed of muon⁴ detectors, which identify and measure the momenta of muons. It employs several detector technologies separated into two categories: Fast-Response Detectors, to quickly select collisions potentially interesting for physics, and Precision Detectors, to provide very precise positions of muons.

³Transition Radiation occurs when a charged particle traverses materials with different refraction index, such as the boundary between two different media [17].

⁴Muons are particles similar to electrons but ≈ 200 times heavier. This allows them to penetrate much deeper into matter, due to less emitted radiation as they decelerate.

In the first group there are:

- Resistive Plate Chambers (RPCs): gaseous detectors surrounding the central region of ATLAS. They are pairs of parallel plates at an electric potential difference, separated by gas.
- Thin Gap Chambers (TGCs): wire-based gas detectors found at the ends of ATLAS, consisting of 30 μm parallel wires in a gas mixture.

Both of these detectors measure the electric signals of particles as they ionize the gas.

Finally, for the role of Precision Detectors, Monitored Drift Tube (MDTs) are able to determine the position of a muon to an accuracy of 100 μm . They are constituted by more than 380 000 aluminum tubes with a wire at the tube's center and filled with a gas mixture. They are stacked up over several layers, to precisely trace the path of each muon.

1.2.4 Magnet System



(a) The Central Solenoid Magnet at its installation, 2004. From [18].



(b) The installation of the eighth and final coil of the Barrel Toroid Magnet, 2005. From [18].

Figure 1.5: Photos of ATLAS magnets during their installation.

Powerful magnets are used to bend the trajectories of charged particles, enabling the measurement of their momentum and charge. In ATLAS this is done with two different types of superconducting magnet systems: toroidal and solenoidal, utilized in the three main sections: Central Solenoid Magnet, Barrel Toroid and End-cap Toroids.

- Central Solenoid Magnet: surrounds the inner detector at the core of ATLAS. It provides a 2 T magnetic field in just 4.5 cm thickness (Figure 1.5a).
- Barrel Toroid: the largest toroidal magnet ever constructed, providing a magnetic field of up to 3.5 T. It is used to measure the momentum of muons.
- Two End-cap Toroids: placed at each end of the experiment, extending the magnetic field to deflect particles escaping the detector along the beam pipe.

1.2.5 ATLAS upgrades

In order to cope with the new experimental challenges of the HL-LHC, as described in Section 1.1.3, the ATLAS detector will undergo a series of upgrades [19], including a new inner tracker and a new timing detector to provide 4D information of the particles created by the collisions.

- Inner Tracker (ITk) [20]: a new all-silicon tracker will replace the Inner Detector. It will be composed of outer Si strip sensors (4 barrel layers and 6 end-cap disks) and inner Si pixel sensors (5 barrel layers

and end-cap rings). The new pixel system will cover an area of about 13 m^2 , a significant increase compared to 1.9 m^2 of the current pixel detector. The pseudorapidity⁵ range will also increase from $\eta = 2.5$ to $\eta = 4$.

- HGTD: a new silicon pixel detector that will provide timing information in the forward region, to improve vertex reconstruction. The design of the detector and its modules are described in more detail in the next chapter (HGTD and LGADs.)

⁵The pseudorapidity is $\eta = -\ln \tan(\theta/2)$ where θ is the polar angle from the z axis, i.e. the beam direction.

2 | HGTD and LGADs

The upcoming High-Luminosity phase has the goal of increasing the integrated luminosity by a factor 10, enhancing the potential of new physics discoveries. However, it will also bring a variety of new complications, most importantly the increase in pileup interactions¹. A new way to mitigate this effect for higher pseudorapidity values will be *"to use high-precision timing information to distinguish between collisions occurring very close in space but well-separated in time"* [12]. This is set to be accomplished by the High Granularity Timing Detector (HGTD). The HGTD will improve the object reconstruction capabilities of ATLAS, complementing the ITk. This, in turn, will help study a variety of processes that rely on the precise identification and measurement of the physics objects produced during $p - p$ collisions, such as jets and leptons. The detector's active elements will be silicon sensors, which will provide precise timing information and will withstand the additional radiation of the HL. To achieve the HGTD's precision goal of 30 ps (50 ps) at start life (end life), the sensors will need to have a time resolution per hit of 35 ps (70 ps). Additionally, a hit efficiency above 95% and a minimum collected charge of 4 fC will also be required.

In this chapter we describe the general design of the HGTD, its benefits to object reconstruction and the special type of silicon sensors (LGAD) that will form its basic components.

2.1 The High Granularity Timing Detector

One of the updates to the ATLAS detector will be the installation of the new ITk, which will improve the tracking² and will extend the pseudorapidity range up to $\eta = 4$. While this extension will improve the reconstruction of physics objects, it will bring some difficulties too.

Event reconstruction relies strongly on the precise assignment of a track to the location of the first collision (primary vertex). Generally, a track is associated to a vertex if the track's origin is compatible with the vertex, this compatibility can be determined with:

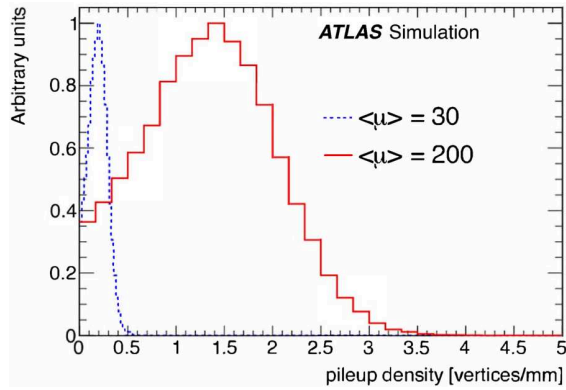
$$\frac{|z_0 - z_{vertex}|}{\sigma_{z_0}} < s. \quad (2.1)$$

Where z_0 is the longitudinal impact parameter, z_{vertex} is the vertex position, σ_{z_0} is the spacial resolution and s is a significant cut (typically 2.5 or 3) [2].

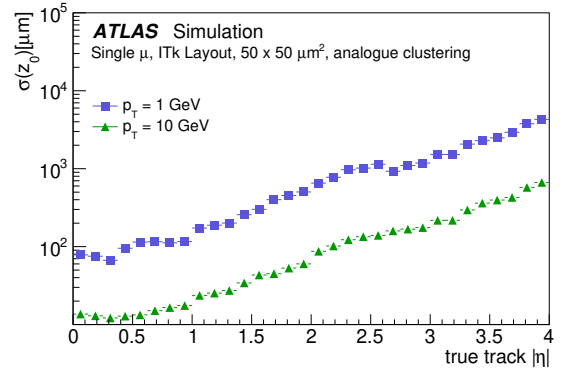
Some problems arise. Firstly, as η increases, tracks become more parallel to the beam and are subject to multiple scattering effects due to more material compared to the central barrel region. Secondly, the HL-LHC will see a rise in the number of collisions per bunch and, as a direct consequence, an increase in the average density of hard-scatter vertices (Figure 2.1a). Lastly, the resolution of the ITk in the z direction (i.e. the

¹Multiple collisions happening during the same bunch crossing, on top of a main event of interest.

²Tracking refers to the process of reconstructing the paths of charged particles as they fly through a detector. From the curvature of the track the momentum and the charge can be inferred. It is a fundamental part of event reconstruction.



(a) Simulation of the local pileup vertex densities for two values of $\langle \mu \rangle$ (average number of collisions per bunch crossing), indicative of the current and future collision rates. From [2].



(b) Simulation of the longitudinal resolution of the ITk as a function of η , for muons with $p_T = 1$ GeV and $p_T = 10$ GeV. From [2].

Figure 2.1: Simulations of the pileup densities (left) and the ITk spacial resolution (right).

beam direction) worsens considerably at higher pseudorapidities, as shown in the simulation in Figure 2.1b for two values of p_T^3 .

The HGTD will provide time information of charged particles with a time resolution of 30ps (up to 50ps at the end of life). This will improve object reconstruction, for example of forward jets and leptons, which is key in many processes, like VBF (vector bosons fusion) and VBS (vector bosons scatter).

The most powerful way to utilize this timing information is to require each track to satisfy:

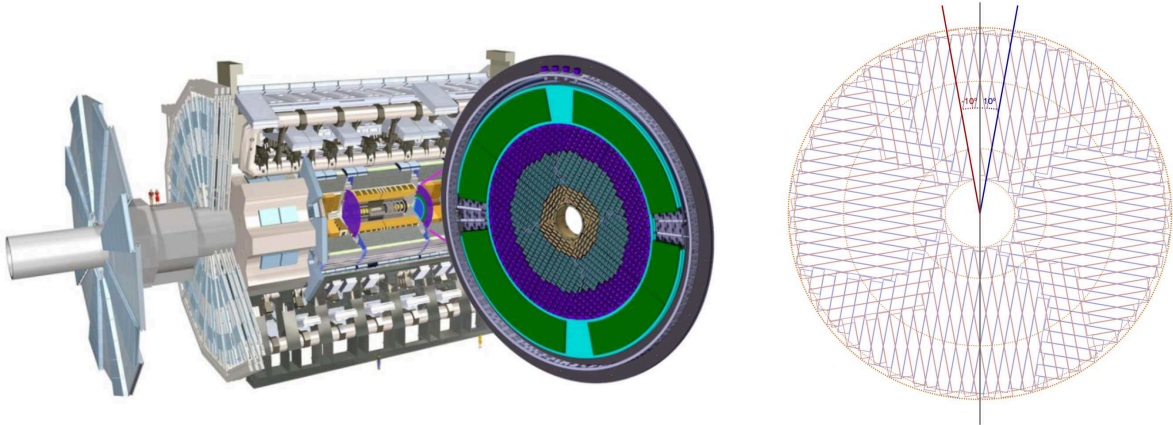
$$\frac{t_{track} - t_0}{\sigma_t} < s. \quad (2.2)$$

Where t_0 is the time of the primary vertex, t_{track} is the time of the track, σ_t is the combined error and s is a significant cut, analogously to Eq. 2.1. The detector will also introduce new timing information to the data, independent of other detector measurements. And finally, the HGTD will provide an additional, precise way to measure luminosity both online (instantaneous, during operations) and offline (calculated after data taking), contributing to the ATLAS goal of 1% uncertainty on luminosity, which is the largest source of uncertainty in many analyses, such as for precision measurements of the Higgs boson [12].

The detector will cover pseudorapidities between $2.4 < \eta < 4.0$ (so an incident angle roughly between 2° and 10°), complementing the ITk (Inner Tracker). The detector consists of two thin disks to be placed outside the ITk, as shown in Figure 2.2a. Each disk includes two instrumented double-sided layers, a hermetic vessel and two moderator pieces (inside and outside the vessel), to reduce the back-scattered neutrons created by the end-cap/forward calorimeters. The detector is further divided into three concentric active regions, as can be seen in Figures 2.2a and 2.2b, ranging: 120 mm-230 mm, 230 mm-470 mm and 470 mm-640 mm [12]. Beyond the outer region are the peripheral electronics. The two disks are covered by 8032 modules in total, each module containing two silicon sensors and two ASICs (Application-Specific Integrated Circuit). Each sensor features 15x15 pads of Low Gain Avalanche Detectors (LGAD), a specific type of silicon sensors.

The radiation that the components will withstand depends strongly on the radius. Due to electronics and sensors specifications, a minimum charge of 4 fC is required. This is achievable up to a radiation damage

³ p_T is the transverse momentum: the component of the momentum perpendicular to the beam line.



(a) The detector will be placed outside the volume of the future Inner Tracker, between the Barrel and the Forward Calorimeters

(b) The overlap between modules on the front and the back was optimized to give an approximately uniform performance (as a function of radius) [12].

Figure 2.2: Location (left) of the HGTD inside ATLAS and orientation (right) of the two faces on each disk. From [2].

level of $2.5 \times 10^{15} \text{ n}_{\text{eq}}\text{cm}^{-2}$ (neutron equivalent⁴); as a result, the sensor and electronics within the smallest ring will be replaced after each 1000 fb^{-1} and the ones within the middle ring will be replaced at half of the data-taking (2000 fb^{-1}).

2.2 Low Gain Avalanche Detectors

A special type of silicon sensors were chosen as the active element for the HGTD. Due to their precision, limited thickness and internal gain.

2.2.1 Working principles of silicon sensors

The core of a silicon sensor consists of a junction between two differently doped layers (Figure 2.3), which means that small concentrations of impurities with either higher atomic number (donors, *n*-type) or lower (acceptors, *p*-type) are introduced inside the crystals. This forms a *pn*-junction, and when a voltage is applied with positive potential on the *n*-side and negative on the *p*-side (reverse bias) the volume between the two layers is depleted of mobile charges, and it becomes an insulator with an internal electric field.

When a charged particle traverses this depletion layer it frees up electron-hole pairs, which move to the electrodes and can be observed as a pulse in the electric potential. Silicon sensors generally offer several advantages over other type of detectors. The lower energy requirement to produce electron-hole pairs leads to improved energy resolution. Moreover, compared to gas detectors for example, silicon sensors have much higher density, allowing precise measurements in significantly thinner layers. However, this type of device has typically higher manufacturing costs and suffers more considerably from radiation damage.

2.2.2 LGADs

A particular type of silicon sensors are Low Gain Avalanche Detectors (LGAD), an illustration is shown in Figure 2.4. The major innovation over standard silicon sensors is an additional *p*-type layer below the *n*+ electrode. This creates a high electric field region which leads to an avalanche effect² of the electrons.

⁴The radiation damage equivalent to neutrons at 1 MeV

²When electrons acquire enough energy they can create new electron-hole pairs ('impact ionization'), which can themselves create new pairs and initialize a multiplication chain that leads to an enhanced signal

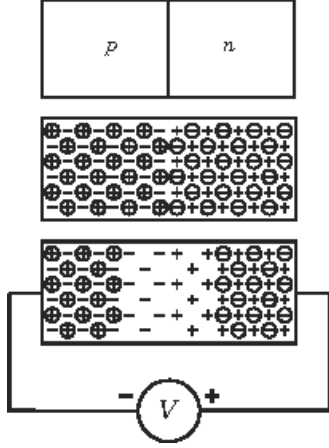


Figure 2.3:

Top: adjacent regions of p -doping (left) and n -doping (right) forming a pn -junction.

Middle: the circled mobile charges (holes for p -type and electrons for n -type) balanced by the charge of atomic cores.

Bottom: when an external (reverse) voltage is applied to the junction, an electric field builds up in the central region, depleting it of free mobile charges [21].

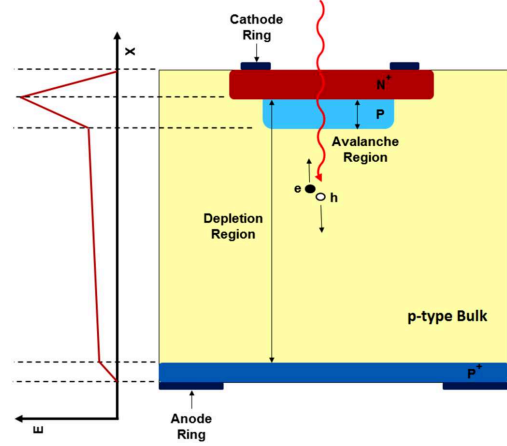


Figure 2.4: Cutaway view of an LGAD, not to scale. The depletion region makes the majority of the volume of the sensor, and above it, lies the avalanche region. On the left of the picture there is a qualitative plot of the electric field along the vertical axis of the sensor, highlighting the strong electric field in the avalanche region. From [2].

Charged particles in standard silicon produce roughly 65 e-h pairs per μm of thickness [22], equivalent to around 0.52 fC for 50 μm . The avalanche effect magnifies the signal around 8-20 times (i.e. a *gain* of 8-20), bringing the collected charge up to 4 fC-10 fC [2].

The intrinsic multiplication in LGADs amplifies the signal, improving the signal-to-noise ratio (S/N) and mitigating the signal degradation caused by radiation-induced noise.

The key metrics of the sensors are: time resolution, collected charge and efficiency.

Time Resolution

To accomplish a time resolution per track of ≈ 30 ps (start) and ≈ 50 ps (end), the sensors will require a time resolution per hit of ≈ 35 ps (start) ≈ 70 ps (end) [12].

Three major effects determine the time resolution of the sensors: time walk from amplitude variations⁵, electronic noise jitter, and Landau fluctuations from charge deposition non-uniformities.

$$\sigma^2 = \sigma_{TimeWalk}^2 + \sigma_{Jitter}^2 + \sigma_{Landau}^2. \quad (2.3)$$

Both time walk and noise jitter depend on the type of readout electronics and depend inversely on the signal slope dV/dt .

- $\sigma_{TimeWalk} = \left[\frac{V_{th}}{S/t_{rise}} \right]_{RMS}$ where V_{th} is the threshold voltage, S is the signal (proportional to the gain) and t_{rise} is the time taken by a signal to go from 10% to 90% of the maximum amplitude.

⁵Time walk refers to the variation in signal timing caused by pulses with different amplitudes reaching a fixed threshold at different times.

- $\sigma_{Jitter} = \frac{N}{dV/dt} \approx \frac{t_{rise}}{S/N}$ where S/N is the signal-over-noise ratio.
- σ_{Landau} depends on the thickness of the sensor (thinner is better) and the setting of the threshold voltage.

It can be noticed that the lowest jitter and time walk errors are achieved with high S/N and small rise time, i.e. thin sensors with high gain. Time walk can also be corrected with reconstruction algorithms, such as constant-fraction-discrimination (CFD), Figure 2.5.

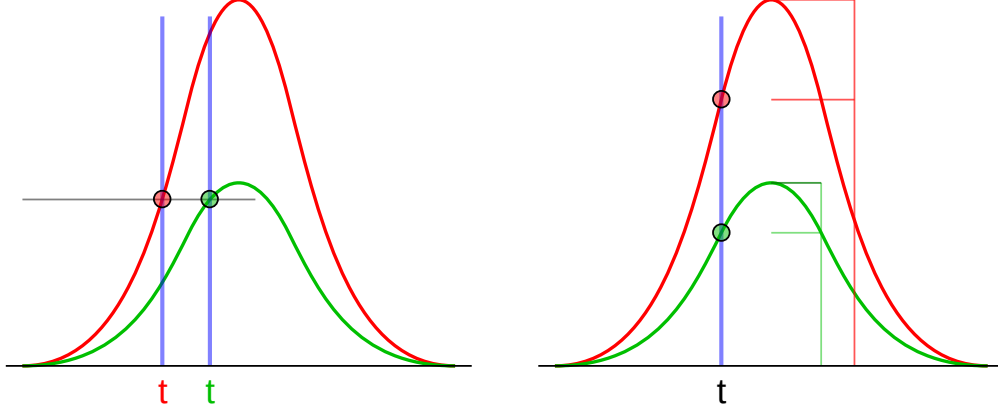


Figure 2.5: Illustration of the *time walk* effect and how it can be mitigated: two pulses with similar shapes, but different amplitudes, can have distinct Time Of Arrivals. Two reconstruction algorithms are compared: Constant Threshold Discriminator (left) and Constant Fraction Discriminator (right). CTD has a fixed threshold value, so the TOA t is set when the pulse crosses the threshold. For CFD, the TOA t corresponds to the moment the pulse reaches a certain fraction of the maximum amplitude, removing the time walk effect.

Radiation damage

Due to the extreme flux of highly energetic particles that the sensors will encounter, they will suffer significant damage. At the microscopic level, this will cause irregularities in the silicon lattice, altering the bulk and surface properties. At the macroscopic level, there will be an increase in leakage current, a degradation in charge collection efficiency and a change in the effective doping concentration [23] (as the defects can behave as donors or acceptors). The leakage current can be sufficiently quenched by operating at low temperatures, the charge collection can be improved by applying higher bias voltages. But the last effect has the most detrimental impact on the properties of the sensors, which will deteriorate as the radiation dose increases.

2.2.3 Signal generation by charged particles

The theoretical probability distribution of the energy loss of heavy particles hitting a thin target (such as the active volume of a silicon sensor) has been solved rigorously by Vavilov in [24]. The namesake distribution is a generalization of the Landau distribution, but its evaluation is more difficult, as it is expressed as an integral over some complicated functions [24, Eq.(4)]. Fortunately, it can be approximated by more straightforward functions depending on the value of a parameter κ (for a more detailed explanation see A.1): for $\kappa \rightarrow 0$ the Vavilov distribution can be approximated by a Landau, for $\kappa \gg 1$ by a Gaussian.

For the conditions of the tests studied in this thesis $\kappa \approx 1 \times 10^{-8}$, well below the limit, thus the Landau function is a good approximation. It can be expressed by [25]:

$$\Phi(\lambda) = \frac{1}{\pi} \int_0^\infty e^{-\lambda u} u^{-u} \sin(\pi u) du. \quad (2.4)$$

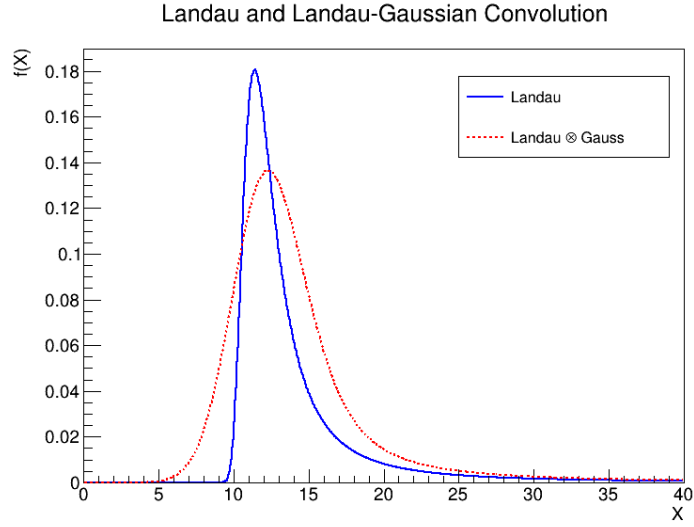


Figure 2.6: Plot of Landau function (in blue) and the Landau x Gaussian convolution (in red), with equal Landau parameters and arbitrary units, both normalized to 1.

Additionally, due to a combination of theoretical corrections [26] (due to the atomic binding of the electrons) and statistical errors on the measurements, the final distribution of energy deposit follows more accurately the convolution of a Gaussian and a Landau distributions (Figure 2.6).

3 | Test Beam setup

A typical way to test the performance of sensors for High-Energy Physics applications is with test beams: the Devices Under Test (DUTs) are put through a focused flux of high energy particles and their response is read out and analyzed.

The HGTD collaboration has run many test beam campaigns, mainly at CERN and at DESY (Deutsches Elektronen-Synchrotron). The focus of this analysis is the test beam campaign of May 2023, conducted at CERN using a beam of 120 GeV pions, provided by SPS.

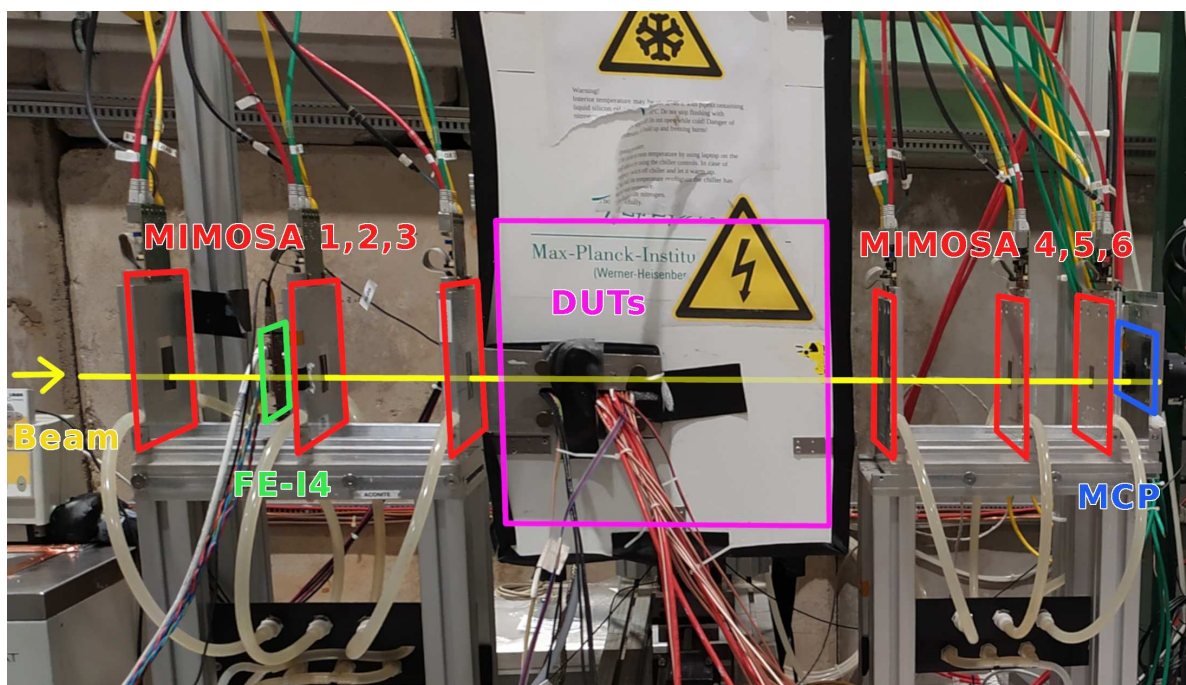


Figure 3.1: The main components of the test beam setup: the EUDET-type beam telescope with six MIMOSA planes, the FE-I4, the MCP and the cooling box containing the DUTs

The experimental setup of this campaign included:

- FE-i4 [27]: a hybrid pixel detector that provided a user-defined Region Of Interest (ROI), creating a HitOR¹ signal.
- Trigger Logic Unit (TLU): provided a trigger², combining the signals from a scintillator and the FE-i4.

¹A binary signal generated when at least one pixel registers a hit

²A trigger is a signal that initiates data readout when a particle is detected

- EUDET-type beam telescope equipped with 6 MIMOSA planes [28]: used for precise spatial measurements of the particles in the beam, enabling track reconstruction.
- MicroChannel Plate detector (MCP) [29]: measured the time of the moving particles, used as reference for the time resolution.
- Cooling box: contained the various DUTs, maintaining them at -30°C .
- 2 oscilloscopes: acquired the data from the MCP and the DUTs.

3.1 TLU and FEi4

The Trigger Logic Unit provided a trigger whenever a particle was detected. This trigger in turn instructed various components (in this case the oscilloscopes, the MIMOSA planes) to initiate data acquisition. More specifically the TLU took input from a scintillator and the FE-i4 to create a trigger.

The FE-i4 (Front-End-Iteration 4) [27] is a pixel readout chip with a planar silicon sensor bump bonded to it. It has a configurable sensitive area of up to $16.8 \times 20 \text{ mm}^2$ made up of 80 columns ($250 \mu\text{m}$ pitch) and 336 rows ($50 \mu\text{m}$).

Selecting a specific ROI allowed to exclude the areas outside the DUTs surface and to reduce sources of noise (e.g. noisy pixels of MIMOSA planes).

3.2 EUDET-type beam telescope

A EUDET-type beam telescope [28] equipped with a total of 6 Minimum Ionizing MOS Active Pixel Sensor planes [30] provided precise spatial measurements of particle trajectories. Each plane consists of 576 rows and 1152 columns of pixels, each sized $18.4 \mu\text{m} \times 18.4 \mu\text{m}$, covering an active area of around $21.2 \times 10.6 \mu\text{m}^2$. Three of the planes were located before the DUTs, in the direction of the beam, and three were located after. The hits provided by the planes enabled the reconstruction of the tracks of the particles in the beam.

3.3 MCP

The microchannel plate (MCP) detector [29] consists of a single block of resistive material with numerous evenly spaced small tubes (microchannels) connecting one face to the other. The main working principle is similar to that of an electron multiplier. When a voltage is applied between the two sides a potential gradient is established along the channels. Whenever a charged particle hits the inner wall of the tubes multiple secondary electrons are emitted, these electrons then are accelerated and hit the opposite wall in the channel, causing the emission of further secondary electrons. As a result, an exponentially increasing number of electrons can be extracted from the output. This detector can provide very fast response time and, for this reason, it was used as a reference for the time resolution of the other devices.

3.3.1 Time resolution of the MCP

The time resolution of the MCP was first calculated by comparing the time difference of two independent sensors (CNM-W4 and CNM-W5, described in Section 3.5) and the MCP. In this way, it was possible to build a system of three equations of the time differences.

$$\begin{cases} t_{1-2} = t_1 - t_2 \\ t_{1-MCP} = t_1 - t_{MCP} \\ t_{2-MCP} = t_2 - t_{MCP} \end{cases} \quad (3.1)$$

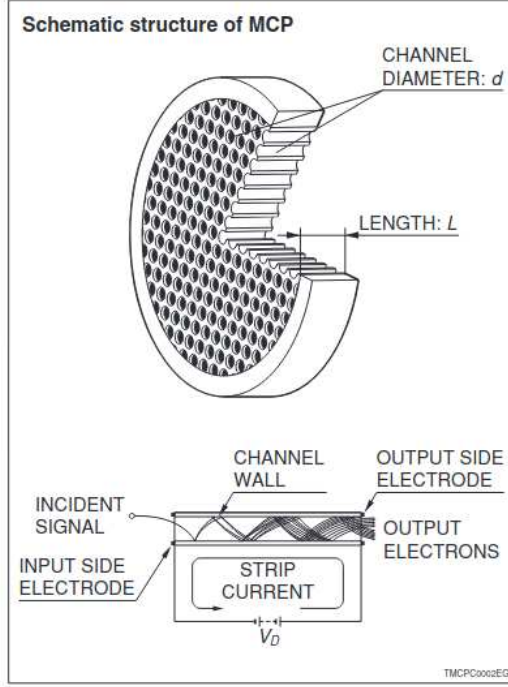


Figure 3.2: Schematics of an MCP.

Top: structure of the detector with a cross-sectional view of the channels, highlighting the diameter d and the length L of the micro-channels. Standard MCPs are fabricated with a diameter of 10s μm and a ratio α ($\alpha = L/d$) 40 to 60.

Bottom: when a voltage (V_D) is applied, the channel walls behave as continuous electron multiplier: when a target object (electron, ion or photon) hits the inner walls, electrons are released in a parabolic trajectory, and they start a chain reaction which produces a signal amplified by $10^4 - 10^7$ times. From [31].

The time distributions (left sides of the equations) were fitted with a Gaussian function, each with a width $\sigma_{ij} = \sigma_i \oplus \sigma_j$, where i and j are two of the devices in Equation 3.1 (i.e. MCP, CNM-W4 and CNM-W5). Assuming that the devices were all independent, their time resolutions would also be independent. This gave a system of three equations with three unknowns, which could be solved analytically to find each individual time resolution: $\sigma_1, \sigma_2, \sigma_{MCP}$.

This procedure was repeated for all available runs, and the results computed this way are shown in Table 3.1.

Table 3.1: Time resolution values of the MCP. The uncertainty is the standard deviation of all the computed runs.

Voltage [V]	2500	2600	2800
Time resolution [ps]	36.52	16.48	3.73
Uncertainty [ps]	± 0.81	± 0.57	± 1.33

3.4 Oscilloscopes

The signal generated by the DUT was recorded by two four-channel oscilloscopes: channel 1 (denoted as Ch1) of both oscilloscopes was connected to the MCP (whose signal was put through a splitter and sent to both oscilloscopes). Therefore, up to 6 channels were available in each batch to be attached to the DUTs. The waveforms recorded by the oscilloscopes were then processed with the PyAna software [32].

3.5 The sensors

A variety of LGADs were tested: with different manufacturers, different designs and different fluence³. They were divided into three main categories:

³Radiant exposure, here measured in neutron-equivalent over square centimeters

- USTC: from the University of Science and Technology of China.
- IHEP: from the Institute of High Energy Physics, China.
- CNM: from the Centro Nacional de Microelectrónica, Barcelona, Spain.

The first two designs have both been manufactured by the IME (Institute of Microelectronics) of the Chinese Academy of Sciences. The two unirradiated CNM sensors (labeled CNM-W4 and CNM-W5) were used to measure the time resolution of another device, the MCP (Section 3.3), which was used as a time reference for the time measurements of all the other LGADs.

Two other devices from CNM were irradiated at $1.5 \times 10^{15} \text{ n}_{\text{eq}}$ and $2.5 \times 10^{15} \text{ n}_{\text{eq}}$. Two versions (v2 and v3) of the IME-IHEP sensors were available, both had a carbon implantation, which has the purpose of increasing the radiation hardness of the LGAD. The sensors were either a single pad or an array (2×2 or 1×3) of pads, and they were irradiated at fluences ranging from 0 to $2.5 \times 10^{15} \text{ n}_{\text{eq}}$. Two USTC devices, also with carbon enrichment (version 2.1), were tested at zero and low fluence.

Notably, the data from the irradiated USTC sensor was not available. Moreover, one pad of IMEv3-W12-2x2-1.5E15 was removed from the analysis due a possible wrong labeling (see in Appendix A.3).

The sensors were then installed onto readout boards. In the case of multi-pads sensors, some number of pads were each connected to one channel of the board, so that they could be measured individually.

Table 3.2 provides the list of the devices that were characterized.

Table 3.2: List of the tested devices.

Device name	Vendor	Pads, used channels	Fluence [$n_{\text{eq}}/\text{cm}^2$]	Radiation type ⁴	Notes
CNM-W4	CNM	single	0	-	reference
CNM-W5	CNM	single	0	-	reference
CNM-W5-1.5E15	CNM	single	1.50×10^{15}	neutron	-
CNM-W3-2.5E15	CNM	single	2.50×10^{15}	neutron	-
USTC2.1-W17	USTC	2×2 , 2 channels	0	-	-
USTC2.1-W17-2E14	USTC	2×2 , 1 channel	0	-	not available
IMEv3-W12-2x2	IHEP	2×2 , 2 channels	0	-	
IMEv3-W12-1x3	IHEP	1×3 , 2 channels	0	-	
IMEv3-W12-2x2-1.5E15	IHEP	2×2 , 3 channels	1.50×10^{15}	neutron	only two channels
IMEv2-W7-1E14	IHEP	single	1.00×10^{14}	proton	-
IMEv2-W7-6.5E14	IHEP	single	6.50×10^{14}	proton	-
IMEv3-W16-8E14	IHEP	single	8.00×10^{14}	proton	-
IMEv3-W16-1x3-1.5E15	IHEP	1×3 , 1 channel	1.50×10^{15}	neutron	no usable batches
IMEv3-W16-2.5E15	IHEP	single	2.50×10^{15}	neutron	-

⁴ As protons and neutrons interact differently with the atoms of the silicon lattice, they can introduce distinct defect structures. In addition, the enrichment of various impurities may lead to further differences in the type of damage [33].

4 | Analysis

The data recorded by the oscilloscopes, in the form of waveforms, was pre-processed using the software PyAna [32]. Another software, PaTrack [34], used the spacial information from the MIMOSA planes to reconstruct the tracks of the particles. The data was then combined into ROOT files to be further analyzed.

Some quality cuts were applied to the data, to remove extra noise. To illustrate the analysis procedure that was followed in this study, a single DUT was picked as an example. All the plots in this chapter (unless otherwise stated) refer to the sensor IMEv3-W12, a 2×2 array of LGADs, operated at -30°C and perpendicular to the beam.

4.1 Data pre-processing

The pre-processing was performed by other collaborators prior to this analysis. The steps of the pre-processing are described below.

Whenever a trigger was detected, the oscilloscopes recorded an event. For each event, at least 2000 samples were collected, spaced every 25 ps, an example of one event (a pulse) is shown in Figure 4.1. This pulse (waveform) was then processed with PyAna.

Firstly, the pedestal and the noise were measured by computing the mean and standard deviation, respectively, using the first 240 samples, where no signal is expected [35]. Secondly, the maximum of the pulse was found by a second-degree polynomial fit, in a 400 ps window around the sample with the highest amplitude. After subtracting the pedestal, the total charge was computed as the integral of the pulse divided by the transimpedance¹ value of the board the LGAD is mounted on (Equation 4.1). The integral is computed in a window wide enough to contain the full pulse.

$$Q = \frac{\int_{t_1}^{t_2} V(t)dt}{R_b} . \quad (4.1)$$

Where Q is the charge, V is the voltage (as a function of time t), R_b is the transimpedance and $t_2 - t_1$ is the integration window.

The spacial information was processed with the PaTrack software to create tracks of the particles. Two straight tracklets, an "upstream" and a "downstream" tracklet passing through MIMOSA planes 1,2,3 and 4,5,6, respectively, were reconstructed to best fit the hits on each plane and meet at the longitudinal (z) location of the DUTs, Figure 4.2. This provided each event with the XY intersection of the track onto the plane of each DUT.

To measure the time resolution it is crucial to define the Time of Arrival (TOA), i.e. the time at which the particle hits the sensor and the pulse is recorded. There are several ways to make this choice, includ-

¹Transfer impedance, it is the ratio of output voltage to input current in the readout board, expressed in Ω

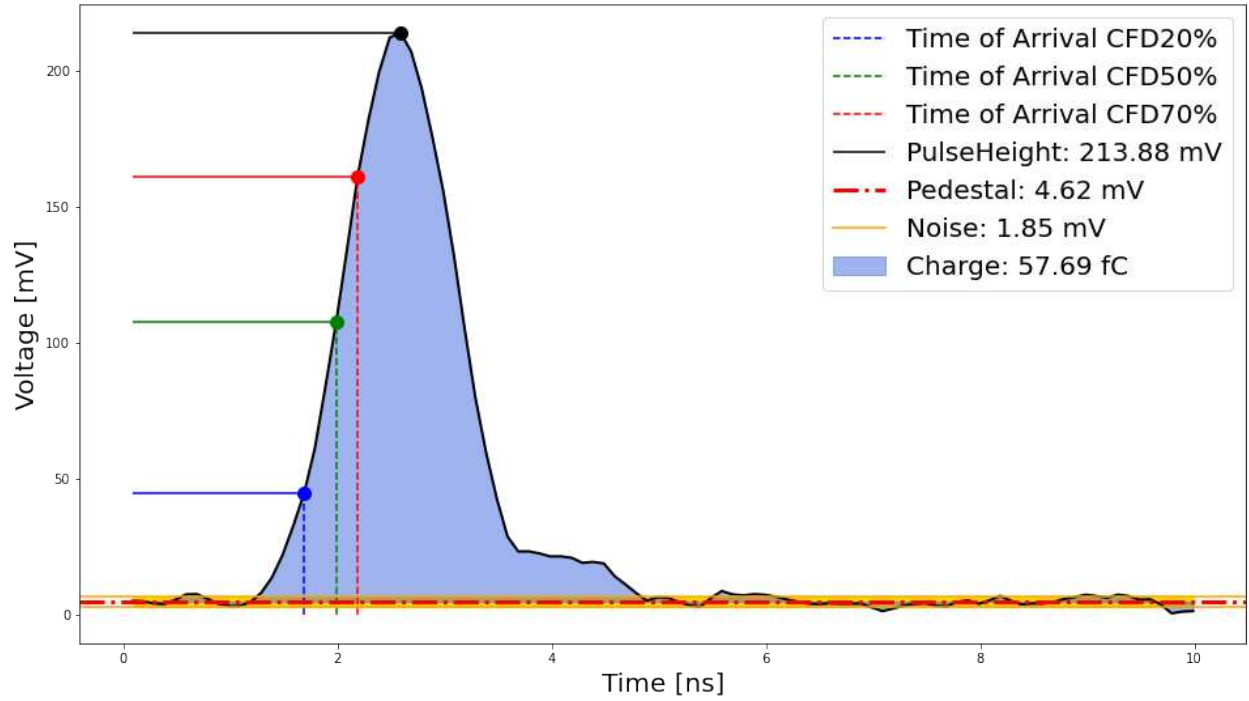


Figure 4.1: An example of a pulse with its main features highlighted. The pedestal (red dotted line) is the mean of the background signals, the noise (yellow) is its standard deviation, the pulse height (black) is the maximum amplitude and the charge (light blue) is the area under the pulse. The remaining three dashed lines (blue, green and red) correspond to the TOA associated to three different values of CFD (20%, 50% and 70%, respectively).

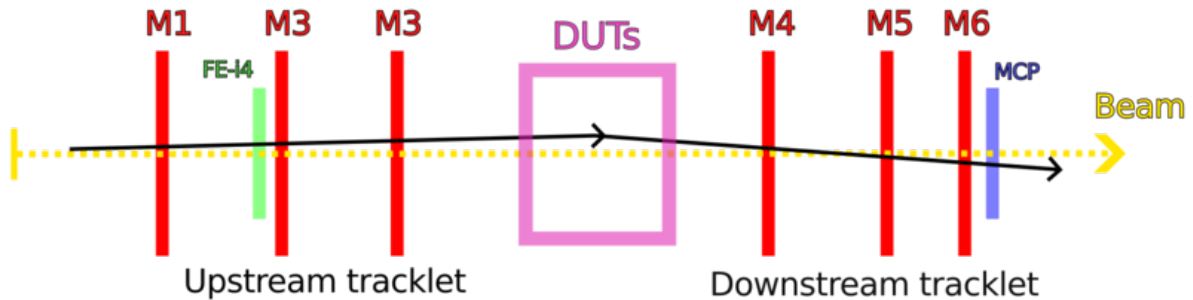


Figure 4.2: Scheme of the reconstructed tracks, made of two tracklets, one upstream and one upstream, as they best fit the hits on the MIMOSA planes before and after the DUTs.

ing: Constant Threshold Discriminator (CTD), Constant Fraction Discriminator (CFD) and Zero Crossing Detector (ZCD), Figure 2.5.

In this study we chose to use the Constant Fraction Discriminator. More specifically, for the MCP a value of 50% CFD was chosen, while for the DUTs a value of 20% was chosen in all cases except for heavily irradiated sensors, in which 70% yielded improved results overall. (An example of each in Figures A.4 and A.5.)

4.2 Quality cuts

To ensure that only physically meaningful events were used in the measurements, it was necessary to apply some quality cuts to the initial data. These cuts were designed to reject events affected by various sources of noise or not meaningful to the analysis.

To begin with, we removed events with low signal peak (or high noise). Subsequently, we used the pulse height information to identify the outline of each pad and select only events passing through it. Finally, we established a time window around the peak of the distribution to avoid negative effects caused by the neighbouring pads and the edges of the sensors (outside the gain layer).

4.2.1 Noise cut

The most straightforward trimming consisted in the exclusion of signals which had a pulse height less than 3 times greater than the noise:

$$P > b + 3\sigma.$$

Where P is the maximum of the pulse (pulse height), b is pedestal baseline and σ is the noise. Since the noise is approximately normally distributed, applying this cut ensured that events arising from noise fluctuations were excluded with high confidence.

4.2.2 Pulse height cut

By plotting the distribution of pulse height values, we applied a cut to the pulses located below the local minimum in the distribution², to separate guaranteed signals from potential noise. The goal of this cut was to select only events that were detected by the DUT, and use this information to find the physical outline of the pad.

Using this selection of events, the X and Y coordinates of the tracks were plotted, giving the result shown in Figure 4.5b. The outline of the sensor is distinctly shown, making possible the subsequent quality cut: the selection of tracks that passed through the surface of the pad, i.e. a *geometry cut*

4.2.3 Geometry cut

Looking at the projections of hits on the X and Y coordinates, it was possible to determine the outline of the pad. The distribution was smoothed³ and its gradient (derivative) was calculated. This gave rise to two remarkably pronounced peaks shown in Figure 4.4, which could be interpreted as the edges of the pads. By applying this procedure to both the X and Y projections we determined the horizontal and vertical edges, respectively. The final rectangular shape can be seen in Figure 4.5b, alongside with the density of hits after applying a *pulse height cut*. In few cases the sensors were slightly tilted in the XY plane, giving rise to slightly imprecise outlines, this is shown in A.6, although it did not have a significant impact on the rest of the analysis.

²A Kernel Density Estimator was used in order to smooth the distribution before identifying the minimum

³Using a Kernel Density Estimator, as done previously for the pulse height distribution.

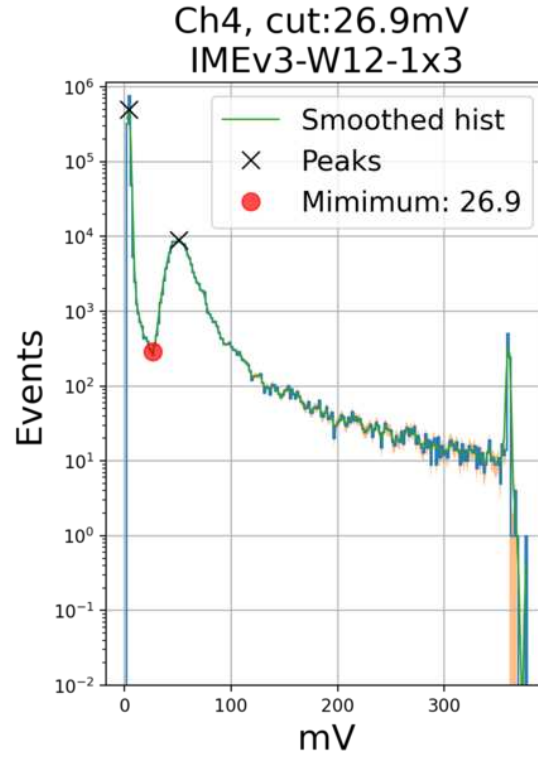


Figure 4.3: The pulse height cut was defined by the position of the local minimum between the two main peaks (identified as noise and signal). The rightmost spike of events was caused by some erroneous oscilloscope settings, explained in Section 5.2.3.

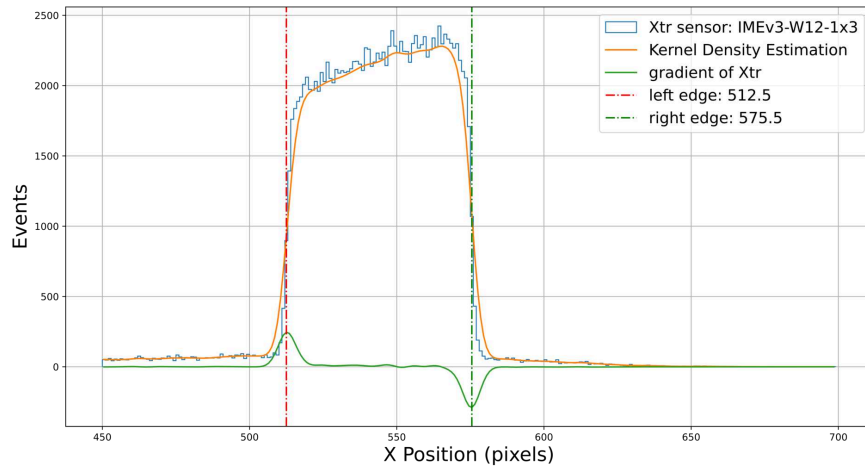
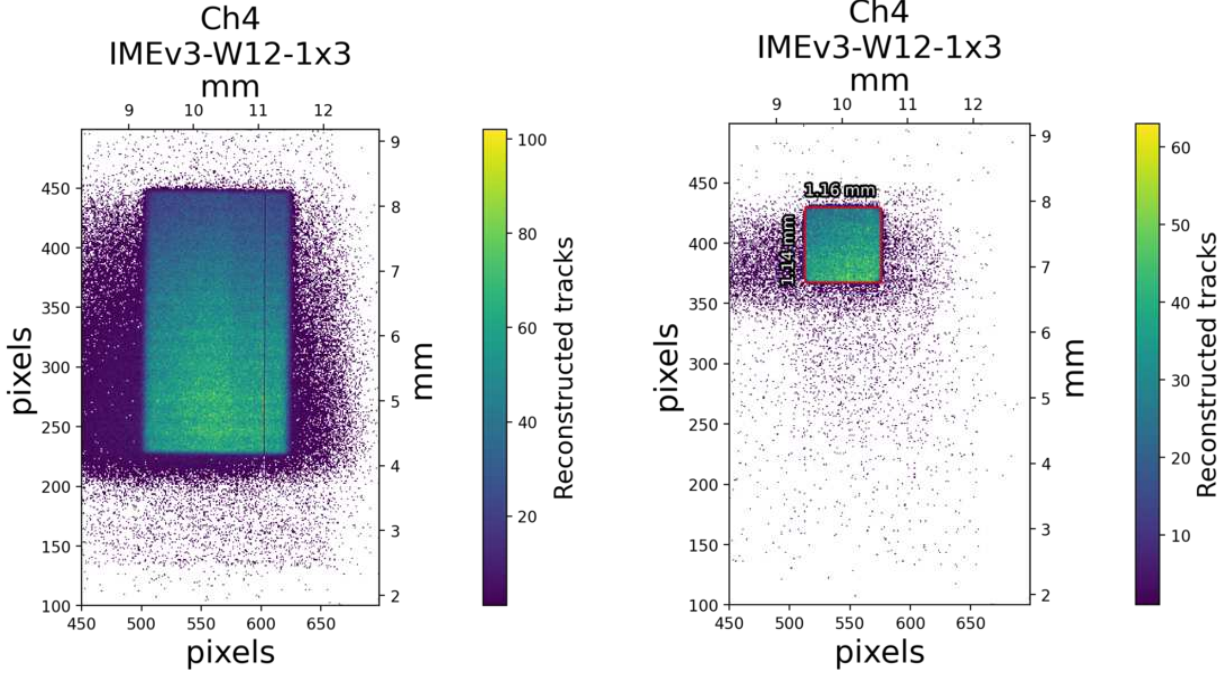


Figure 4.4: Projection of hits on the X axis after the Pulse height cut. By taking its derivative, it was possible to determine the position of the edges of the sensor. The unit is pixels of MIMOSA planes.

For certain properties that were analyzed (such as efficiency and time resolution) it became clear that the rough selection of the contour of the pad was not always sufficient. Thus, an additional *central area cut* was set up. A region of $0.5 \times 0.5 \text{ mm}^2$ in the center of the pad was selected (similarly to what had been done in [36]) in order to avoid undesirable effects originating from the region outside the gain layer of the LGAD.



(a) The heatmap of the reconstructed tracks without any cuts applied. The large rectangular shape is produced by the region of interest (ROI) selected with the FE-I4.

(b) The smaller rectangle corresponds to the area of the pad (highlighted in red with sides lengths) found by applying a *pulse height cut*.

Figure 4.5: The events selected by applying a pulse height cut

4.2.4 Time cut

The physical distance between the MCP and the DUT was fixed, and the momentum of the particles was constant throughout all the runs. Thus, the flight time of the particles between the MCP and the DUT was peaked around one point. Using this knowledge, an additional cut on the Δt was applied.

Firstly, the distribution of Δt ($t_{DUT} - t_{MCP}$, time of arrival of the DUT - MCP) without any other cut was fit with a Gaussian plus a uniform background (Figure 4.6):

$$f(x, A, \mu, \sigma, BG) = A \cdot e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2} + BG.$$

A : Amplitude (n° of events); μ : Mean of the gaussian; σ : Standard deviation; BG : Uniform Background.

Ultimately, a *time cut* was defined as all the events lying inside an interval of a number n of standard deviations σ :

$$(\mu - n\sigma; \mu + n\sigma). \quad (4.2)$$

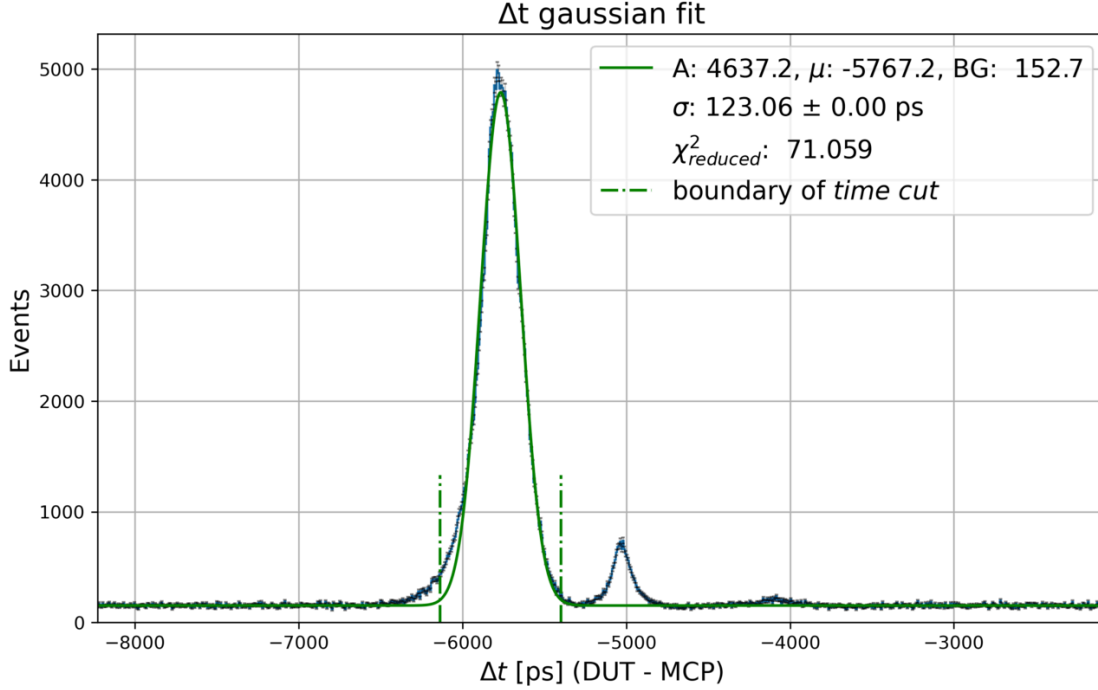


Figure 4.6: Time difference distribution ($t_{DUT} - t_{MCP}$) fit with a Gaussian + uniform background. The vertical dashed lines represent the boundary set as *time cut*. The presence of the secondary peak is explained in Section 5.2.2.

Given that roughly 99.7% of all values of a normal distribution lie within 3 standard deviations, $n = 3$ was deemed an adequate choice.

This cut proved to be especially useful in calculating the total efficiency of each pad and in fitting the charge distribution.

Although, some peculiarities became apparent, namely the second peak shifted to the right of the main one (meaning delayed compared to the latter) and some discrepancy with the expected Gaussian distribution around the left tail. These effects were further investigated and will be explained later in Sections 5.2.2 and 5.2.1.

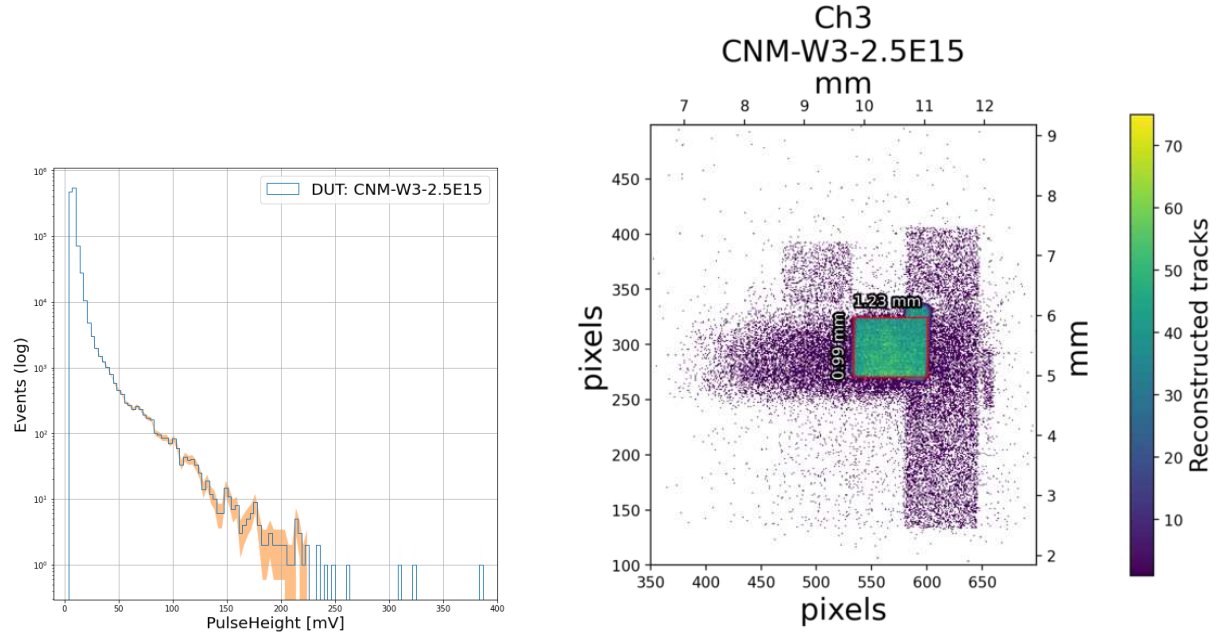
Heavily radiated sensor case

In some cases, typically for heavily radiated sensors, the noise in the pulse height distribution was too large (Figure 4.7a) to allow a *pulse height cut* as in Section 4.2.2. This happened due to the pulse height decreasing and the noise peaks rising, so the separation between the two became less marked. In these situations it was possible to apply a *time cut* instead and still get a satisfactory contour of the pad. The results of the two methods were very similar (Figure A.3 in Appendix) so this seemed a good alternative.

4.2.5 Quality cuts validation

To qualitatively confirm the soundness of the cuts performed, we plotted the density distribution of events of Δt vs **pulse height** (Figures 4.8a and 4.8b). The left plot shows all the available tracks, the right plot shows only the tracks that pass through the smaller central area of the DUT (a square of $0.5 \times 0.5 \text{ mm}^2$ explained in Section 4.2.3).

The narrow selection of events in the right plot is very likely to be real signals (as they are particles guaranteed to have passed through the sensor). By comparing it with the plot on the left it can be noticed



(a) Pulse height distribution, without a clear boundary between noise and signal. The orange belt represents the poissonian error.

(b) Contour of the pad, highlighted in red, found by applying a time cut to the data. The irregular shape of the pad is caused by the ROI selected by the FEi4.

Figure 4.7: Example of a different sensor (irradiated), for which the previous *pulse height cut* became impractical (left), but using *time cut* yielded a reasonable outline of the pad (right)

that a horizontal cut (Pulse height cut) and two vertical cuts (Time cut) provide indeed a good filter for physically meaningful events.

4.3 Collected charge

A central goal of this study was to measure the distribution of the charge collected by the pads and verify their correspondence with the theoretical distribution: a convolution of a Gaussian and a Landau distribution (Appendix A.1).

To achieve this, all the quality cuts defined in Section 4.2 were applied, and a fit was carried out using an implementation of the Gaussian*Landau convolution provided by the ROOT framework [37] (Figure 4.9).

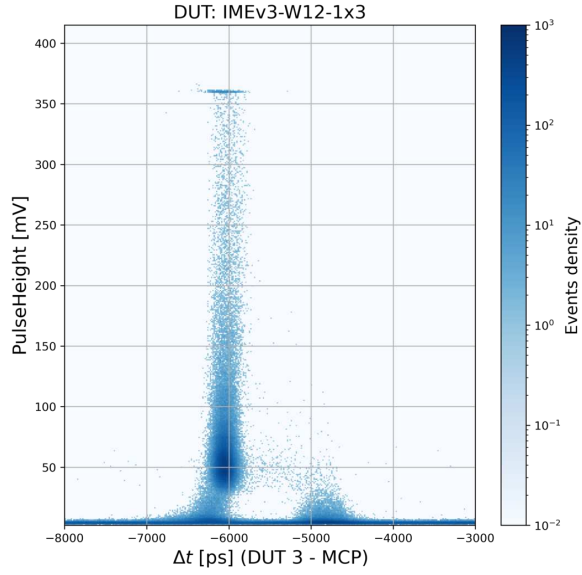
The collected charge of a sensor is defined as the most probable value (MPV) of the charge, i.e. the highest point in the distribution.

Gain

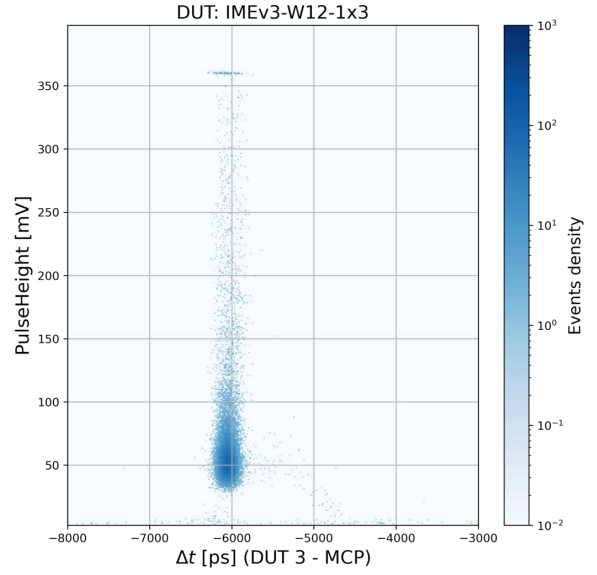
The gain is obtained by dividing the most probable value of the collected charge by the expected charge from a minimum ionizing particle (MIP) in a silicon sensor without gain. For a $50\mu\text{m}$ thick sensor, this value is 3280 e-h pairs, equivalent to 0.52 fC [22].

Extra noise

Despite all the trimming to the data, a noticeable part of the noise centered at ≈ 0 could not be removed. We could not pinpoint the origin of this extra noise, but a lot of factors could contribute to it: track reconstruction, other electronic noise etc. Considering this point, the fit was performed in a smaller interval,



(a) Density plot of all the events without any cuts applied.



(b) Density plot selecting only tracks passing through the central $0.5 \times 0.5 \text{ mm}^2$ area of the pad.

Figure 4.8: Density plot of events when no cuts are applied (left) and when only a central square in the sensor is selected (right). The comparison between the two plots validated our choices made for quality cuts.

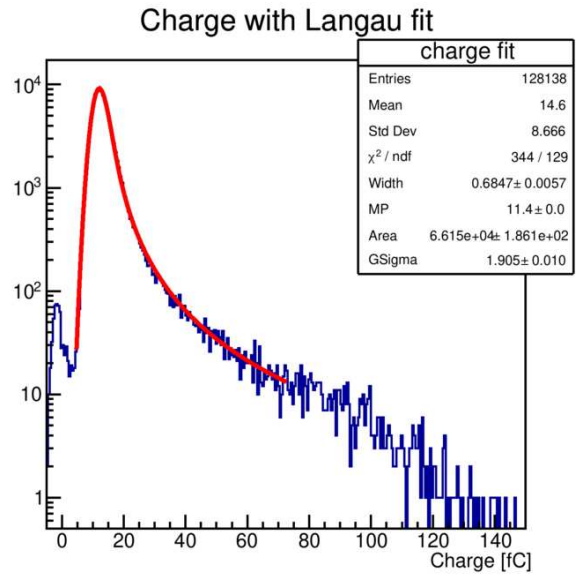
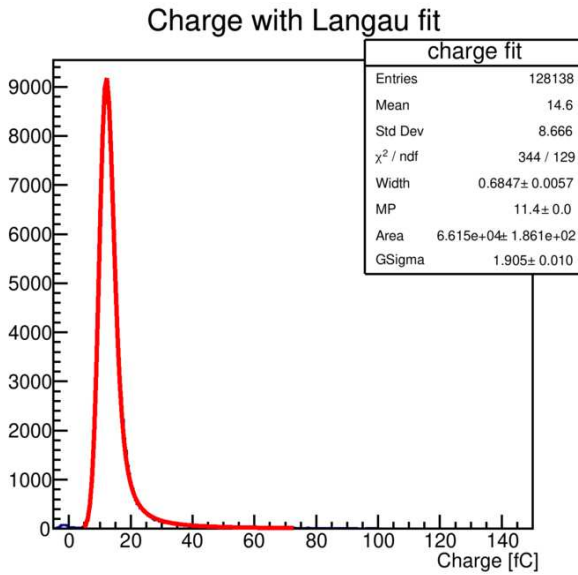


Figure 4.9: Fit of the charge distribution (red) carried out with ROOT. On the right it is shown the same plot in logarithmic scale, to highlight the extra noise at ≈ 0 .

avoiding the noise (Figure 4.9). The left limit was fixed at 4fC, as this is also the lowest signal that the electronic board is able to measure, the right limit was adjusted to include only bins with significant amount of data (and account for the Irregularities in the charge distribution, described in Appendix).

4.4 Efficiency

The efficiency (ϵ) was defined as the total number of tracks with charge larger than a certain *threshold charge* divided by the total number of reconstructed track (**after** quality cuts had been applied).

$$\epsilon = \frac{\text{Tracks with } q > Q_{\text{threshold}}}{\text{Total tracks}}.$$

The threshold charge $Q_{\text{threshold}}$ was chosen to be 4fC, which corresponds to the limit of sensitivity of the ASIC.

Before measuring the overall efficiency of the sensors, we applied a *time cut* (as described in Time cut). Additionally, we elected to restrict the calculation to the smaller *central area* of $0.5 \times 0.5 \text{mm}^2$ defined earlier in Section 4.2.3. This area can be seen highlighted in red in Figure 4.10, which also shows the efficiency value computed for each individual squared binning of the surface.

Finally, the total efficiency was also computed with different values of *threshold charge* to obtain the plot in Figure 4.11, which shows that (in this example) the value remains constant well after the chosen threshold of 4fC.

4.5 Time Resolution

The difference of times of arrival between the DUT and the MCP was fitted with a normal distribution. The standard deviation of said distribution corresponded to the time resolution of the DUT and MCP combined. Using simple error propagation, the time resolution of the DUT was computed:

$$\begin{aligned} \sigma_{\text{dut}+\text{MCP}} &= \sigma_{\text{dut}} \oplus \sigma_{\text{MCP}} \\ &\downarrow \\ \sigma_{\text{dut}+\text{MCP}}^2 &= \sigma_{\text{dut}}^2 + \sigma_{\text{MCP}}^2 \\ \sigma_{\text{dut}} &= \sqrt{\sigma_{\text{dut}+\text{MCP}}^2 - \sigma_{\text{MCP}}^2}. \end{aligned}$$

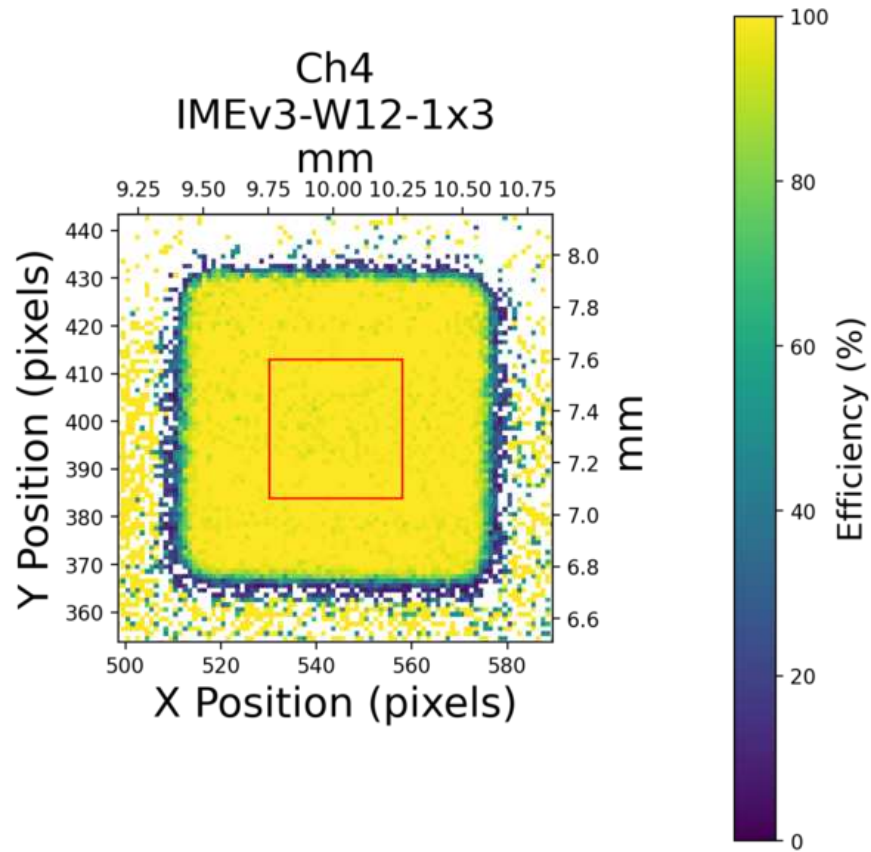


Figure 4.10: 2D histogram plot of the efficiency (per squared bin) and the *central area cut* highlighted in red.

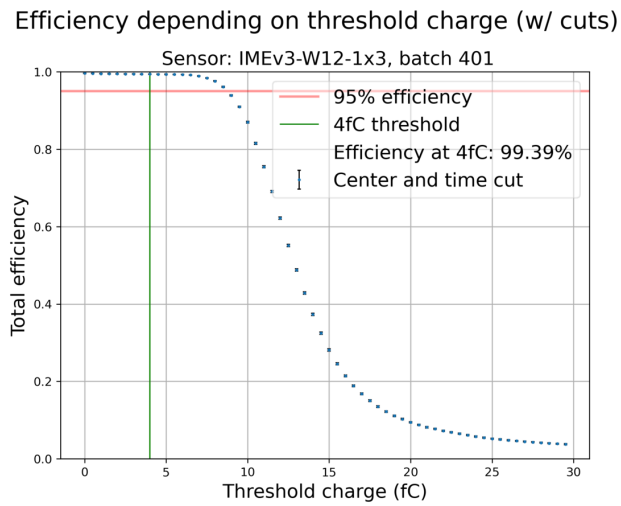


Figure 4.11: Total efficiency when changing the threshold value, ensuring that there is a large plateau of high efficiency.

Time difference, after applying sensor area cut
Batch: 401, Oscilloscope: S1, Ch4

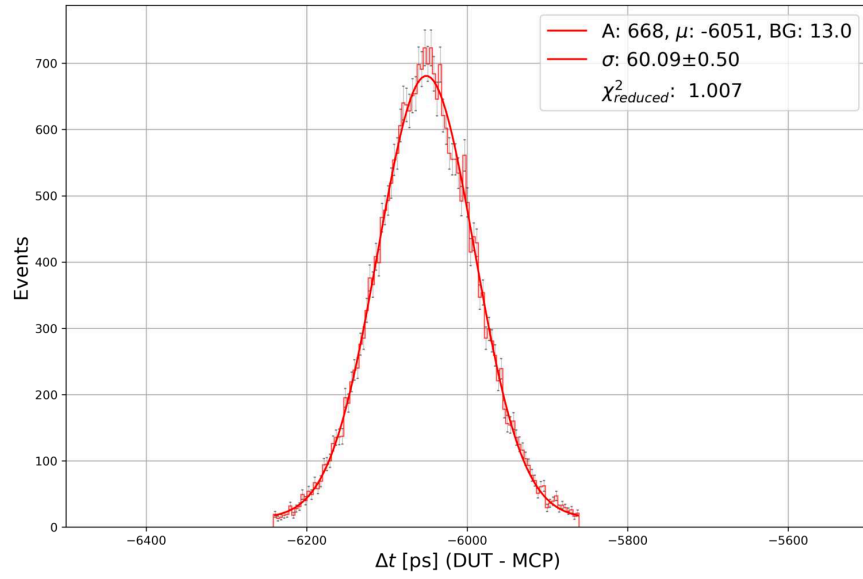


Figure 4.12: Gaussian fit of Δt to find the time resolution (σ), the error bars are the statistical poissonian error ($y_{err} = \sqrt{N}$). For this sensor the final time resolution was $\sigma_{dut} = 47.72 \pm 0.88$ ps.

5 | Results

The main goal of this analysis was to quantify key metrics of the performance of LGAD sensors. In particular, we analyzed the collected charge, the time resolution and the efficiency at the temperature of -30°C , which will be the temperature of the detector in operation. We compared the results between unirradiated and differently irradiated sensors to test whether the requirements at start and end of life were satisfied. The sensors were also studied at angles between 0° and 14° relative to the beam direction, in line with the HGTD's expected pseudorapidity coverage: between 2.4 and 4, which corresponds to incident angles of 2° to 10° .

Additionally, some other properties and effects that arose during the analysis were also investigated. Such as, signal coming from neighbouring pads, noise coming from outside the gain layer of the pads, and properties of the region lying between two adjacent pads. Certain anomalies were also successfully explained, and the root cause was found in the incorrect voltage range set on an oscilloscope, which caused pulses to be "clipped" below a specific threshold. Moreover, due to a temporary malfunction of the cooling box, one batch of data showed neatly the sensitivity of LGADs to temperature variations.

5.1 Main Results

The most important properties of the sensors are:

- Collected charge: the most probable value (MPV) of the Landau*Gaussian convolution (Section 4.3).
- Time resolution: the spread of the Gaussian distribution of the Time of Arrivals (Section 4.5).
- Efficiency: the fraction of incident particles that produce a detectable signal in the sensor. (Section 4.4).

Overall all sensors behaved as expected: their collected charge increased with higher bias voltage and decreased with higher radiation dose (fluence), and most of them satisfied the HGTD's requirements.

5.1.1 Irradiated samples performance

IME

The sensors in the IME group were further split depending on versions and types, to enable more meaningful comparisons between devices with similar characteristics. More specifically:

- IMEv3-W12: 2×2 array unirradiated, 1×3 array unirradiated and 2×2 array at $1.5 \times 10^{15} \text{ n}_{\text{eq}}$.
- IMEv2-W7: single pad at $1 \times 10^{14} \text{ n}_{\text{eq}}$ and single pad at $6.5 \times 10^{14} \text{ n}_{\text{eq}}$.
- IMEv3-W16: single pad at $8 \times 10^{14} \text{ n}_{\text{eq}}$ and single pad at $2.5 \times 10^{15} \text{ n}_{\text{eq}}$.

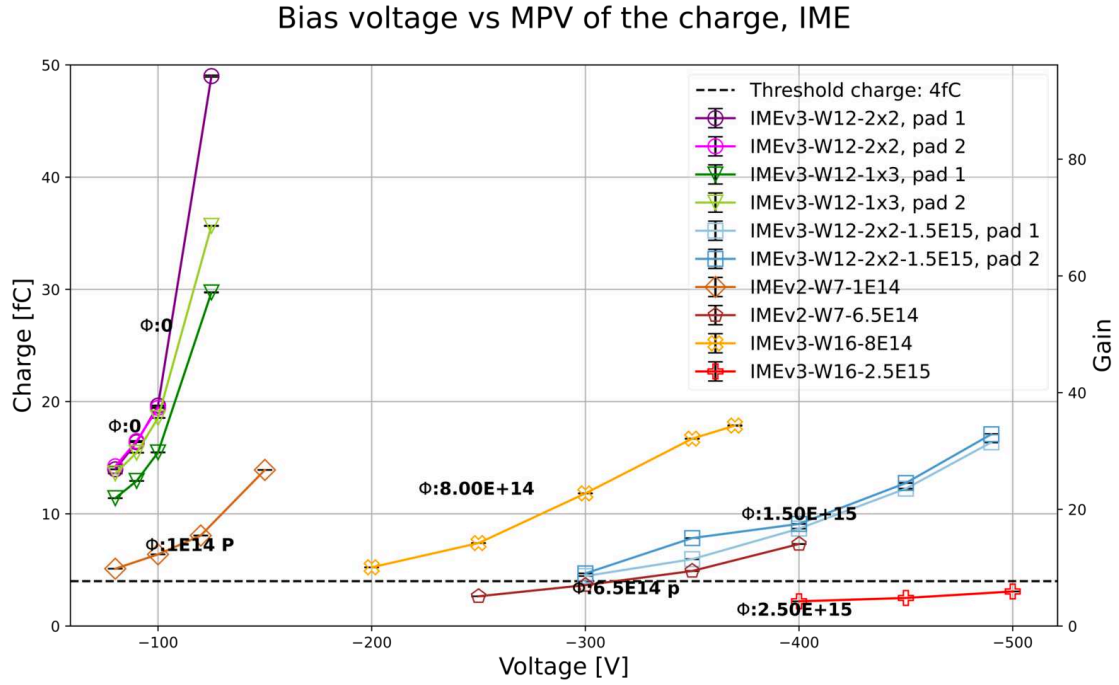


Figure 5.1: Overview of voltage vs collected charge for all the studied sensors from IME. Φ is the radiation dose of each sensor in neutron-equivalent fluence.

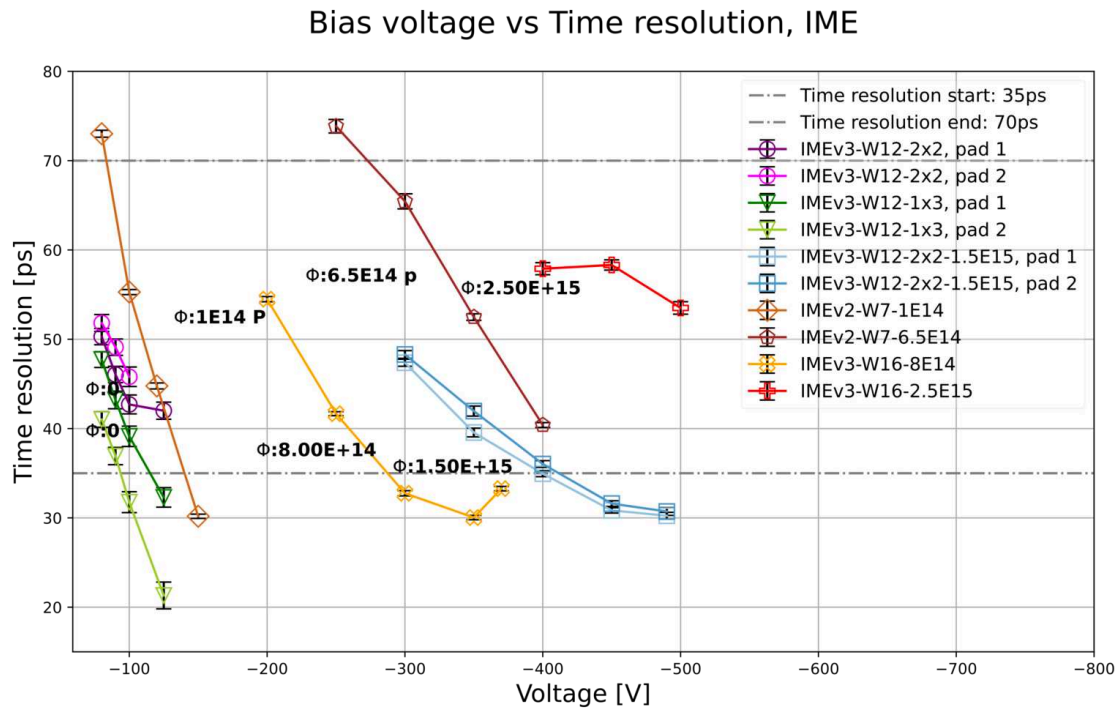
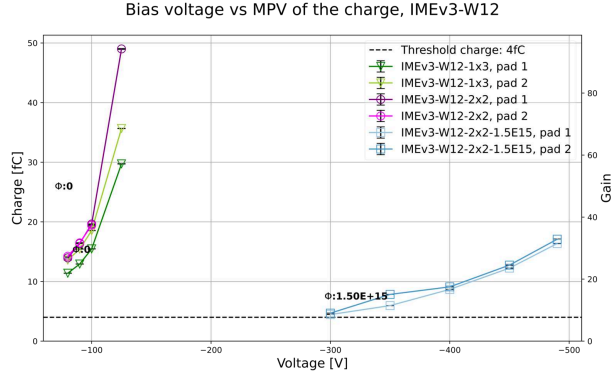
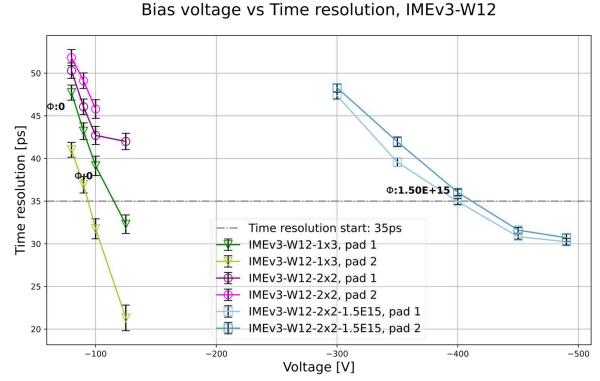


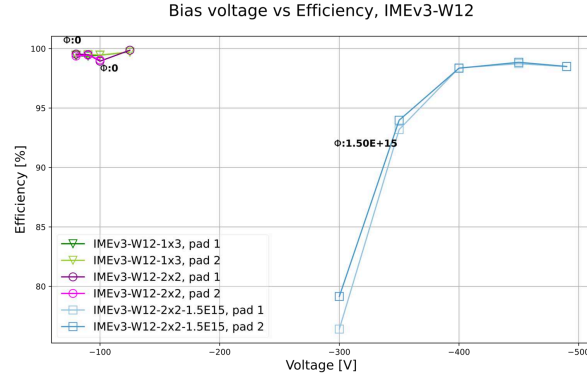
Figure 5.2: Overview of voltage vs time resolution.



(a) Voltage vs Charge.

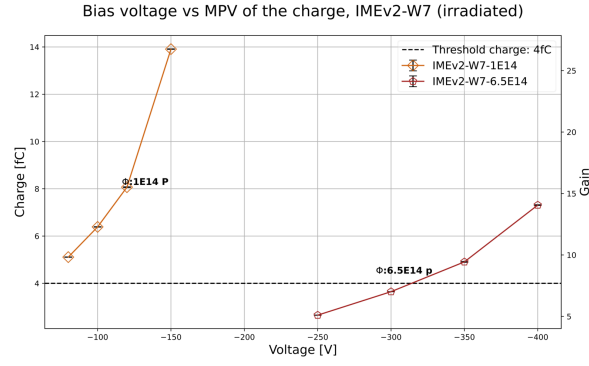


(b) Voltage vs Time resolution

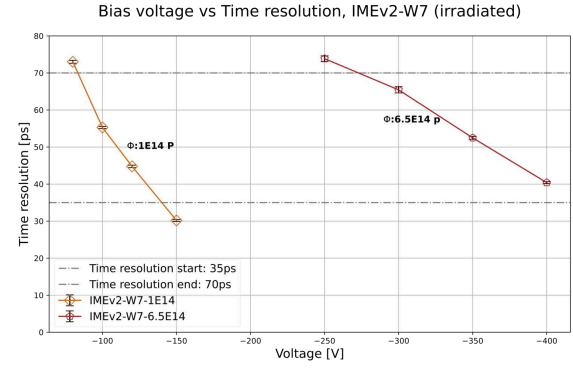


(c) Voltage vs Efficiency

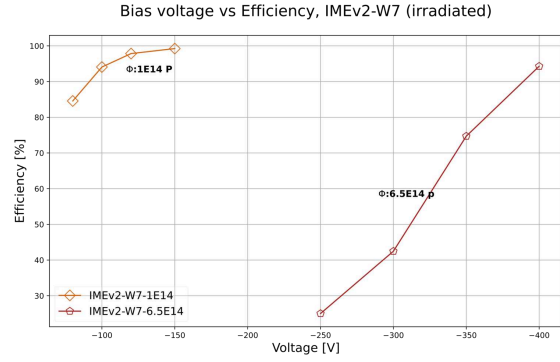
Figure 5.3: IMEv3-W12. The non irradiated sensors (both the 2×2 array and the 1×3 array) had sufficiently high collected charge at all tested bias voltages, while the time resolution was just short of the 35 ps target in the case of the 2×2 array, although this is likely to be solved with higher applied voltage. The lack of data for higher voltages was due to erroneous batches. It should be noted that the two pads of the 1×3 array had significant and consistent discrepancies in charge and time resolution, presumably due to differences at the manufacturing stage. The sensors at $1.5 \times 10^{15} \text{ n}_{\text{eq}}$ had very good performance overall and still reached the target time resolution of beginning of operations (35 ps) at $\simeq 60\%$ of the total expected dose. The efficiency satisfied the 95% goal above $\approx 400 \text{ V}$.



(a) Voltage vs Charge

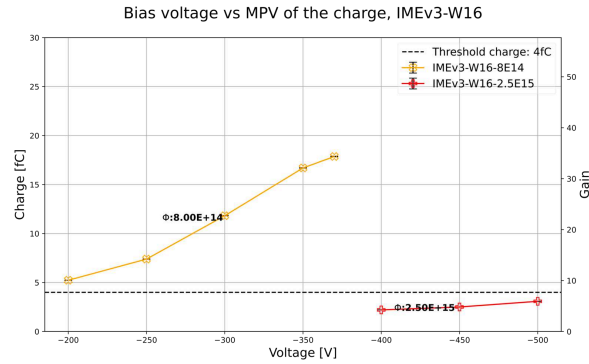


(b) Voltage vs Time resolution

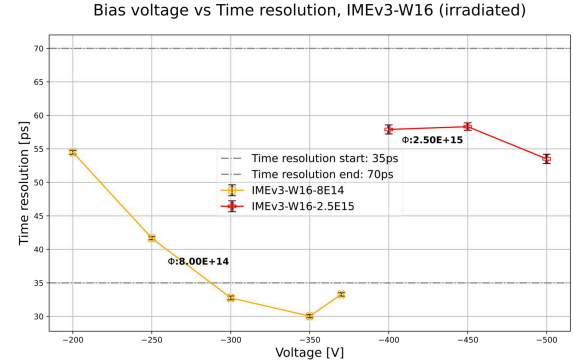


(c) Voltage vs Efficiency

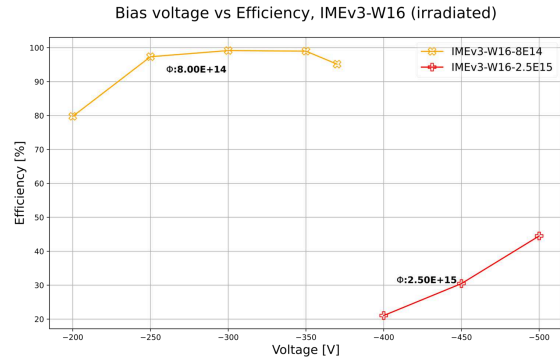
Figure 5.4: IMEv2-W7. For both fluences ($1 \times 10^{14} \text{ n}_{\text{eq}}$ and $6.5 \times 10^{14} \text{ n}_{\text{eq}}$, i.e. low fluence) charge, time resolution and efficiency were within the targets for larger voltages. Although, the more heavily irradiated sensor showed a significant variation in the efficiency, which increased from $\approx 30\%$ up to $\approx 95\%$.



(a) Voltage vs Charge



(b) Voltage vs Time resolution



(c) Voltage vs Efficiency

Figure 5.5: IMEv3-W16. The sensor at $8 \times 10^{14} \text{ n}_{\text{eq}}$ performed well overall, except for the batch at highest bias voltage, which saw a small but unexpected worsening of time resolution and efficiency. We investigated it further, but we were unable to pinpoint the exact cause. The sensor at $2.5 \times 10^{15} \text{ n}_{\text{eq}}$, corresponding to the end-of-life radiation dose, had a collected charge below the requirements, slightly below 3fC and very poor efficiency, but it showed satisfactory time resolution.

USTC

The data from the USTC sensor was the most affected by missing and erroneous batches, as the data from the irradiated sensor was not available.

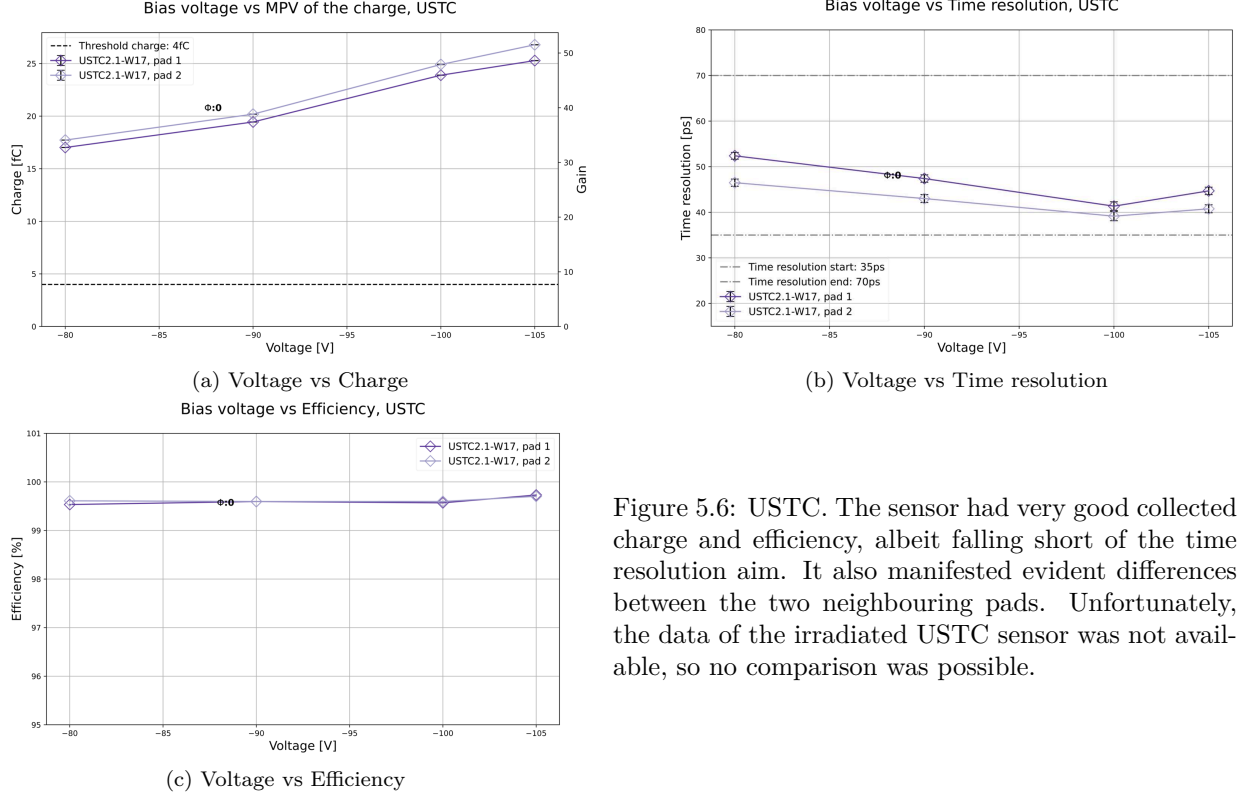
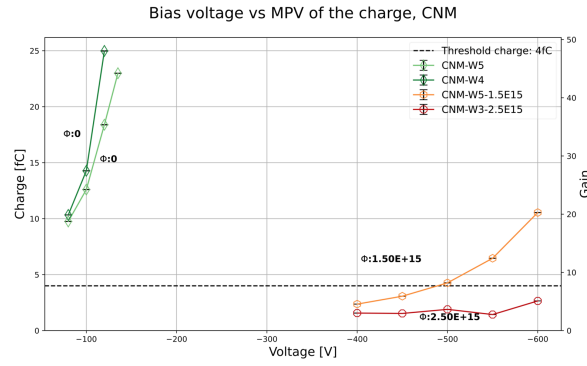


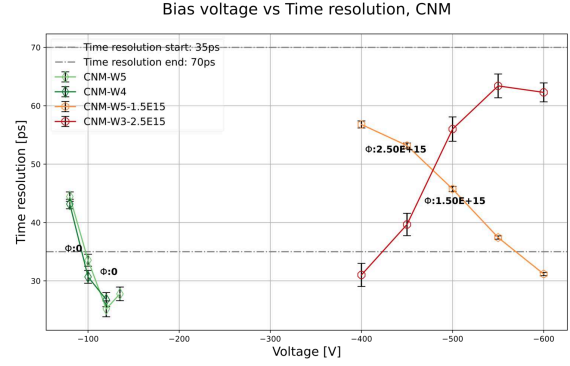
Figure 5.6: USTC. The sensor had very good collected charge and efficiency, albeit falling short of the time resolution aim. It also manifested evident differences between the two neighbouring pads. Unfortunately, the data of the irradiated USTC sensor was not available, so no comparison was possible.

CNM

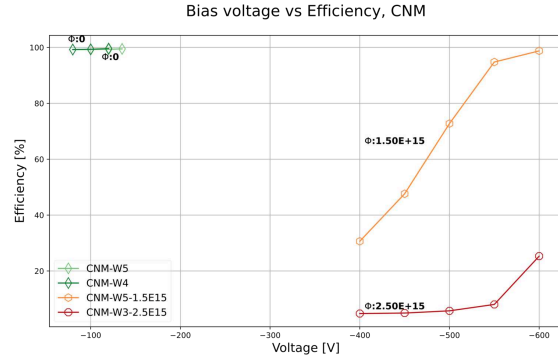
The unirradiated sensors were used to measure the time resolution of the MCP (Section 3.3). They had already been tested before (as in [35]) and they showed similar performance, with gain up to 40 to 50, and time resolution better than 30 ps. In this test beam they were also tested irradiated at two fluences: $1.5 \times 10^{15} \text{ n}_{\text{eq}}$ and $2.5 \times 10^{15} \text{ n}_{\text{eq}}$.



(a) Voltage vs Charge



(b) Voltage vs Time resolution



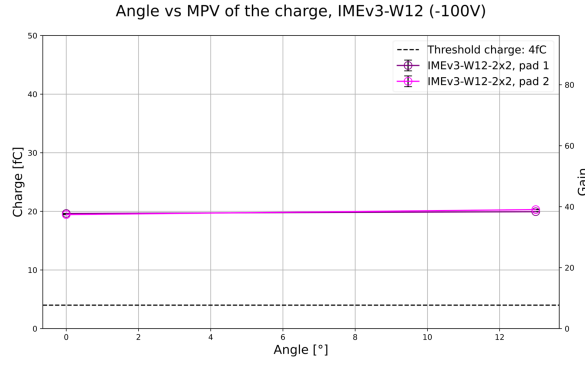
(c) Voltage vs Efficiency

Figure 5.7: CNM. The good performance of the unirradiated sensors was confirmed. At the highest expected radiation dose the sensor had an unexpected counter trend of time resolution, however, the two batches at lower voltages had a significantly lower amount of data available, potentially underestimating the error on the fit. Also, the efficiency (with a threshold of 4 fC) did not reach 30% even at 600 V.

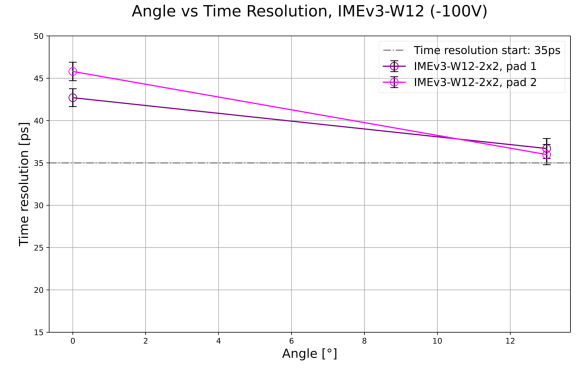
5.1.2 Angled samples performance

When particles traverse a sensor at an angle, the effective path inside the volume of the sensor will be slightly longer ($\approx 3\%$ at 14°). For this reason, charge deposition will be greater and a small improvement in the performance is expected. The sensors were studied with incident angles of 0° , 6° and 14° . However, due to some missing or erroneous batches or mismatched voltages and angles, not all data points were available.

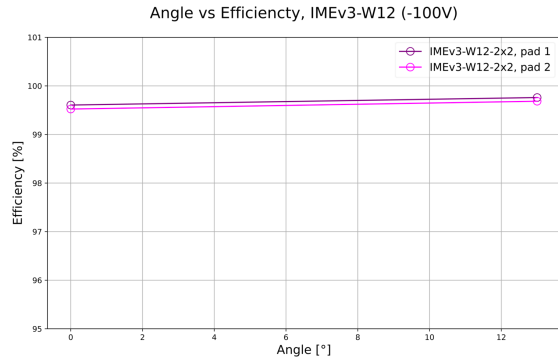
IME



(a) Angle vs Charge

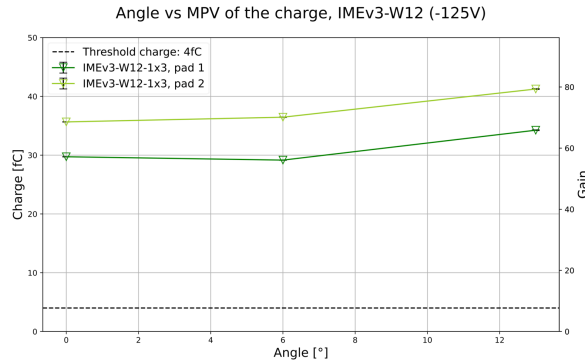


(b) Angle vs Time resolution

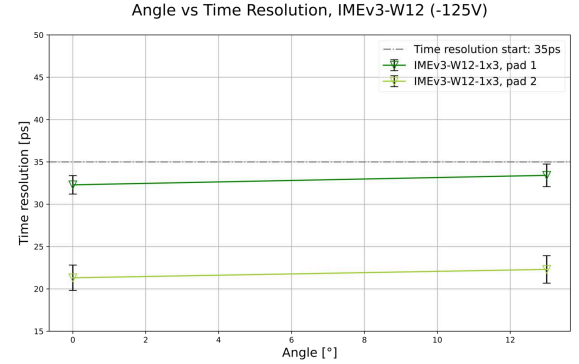


(c) Angle vs Efficiency

Figure 5.8: IMEv3-W12, 2×2 array, at −100 V. The impact of the angle was small, although it showed an improvement in collected charge and time resolution, as expected.



(a) Angle vs Charge

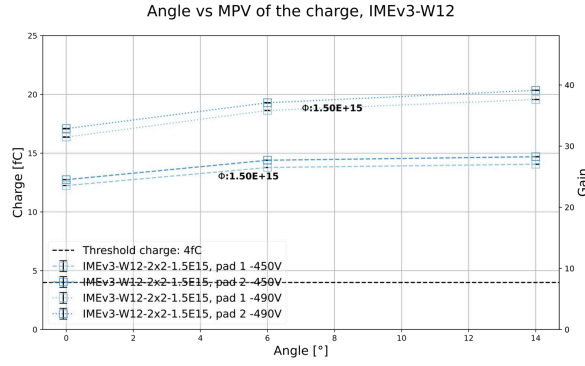


(b) Angle vs Time resolution

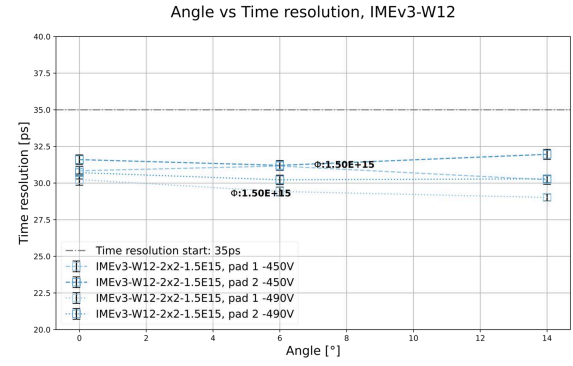


(c) Angle vs Efficiency

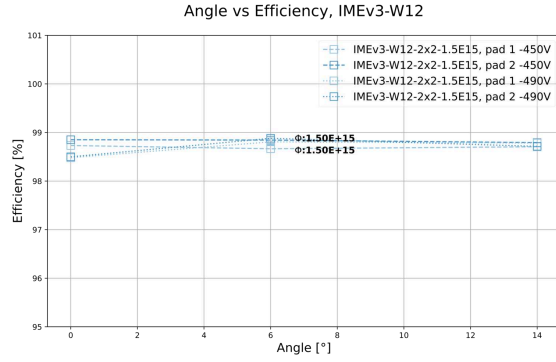
Figure 5.9: IMEv3-W12, 1×3 array, at −125 V. The sensor displayed the predicted increase in collected charge and time resolution, although with significant uncertainty. Furthermore, the differences between the two pads (already seen in Figure 5.3a) were also very evident.



(a) Angle vs Charge

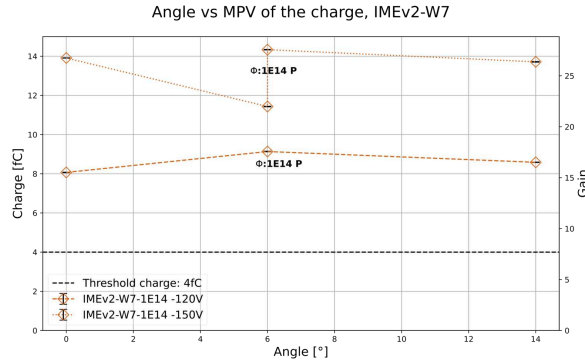


(b) Angle vs Time resolution

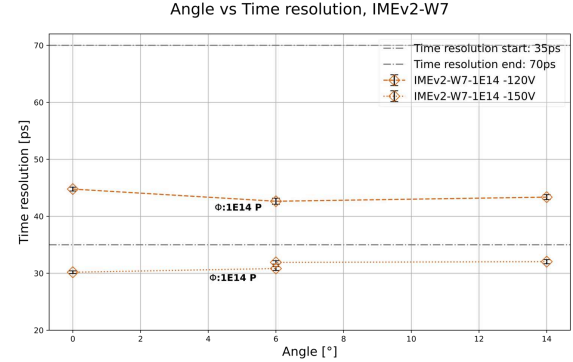


(c) Angle vs Time resolution

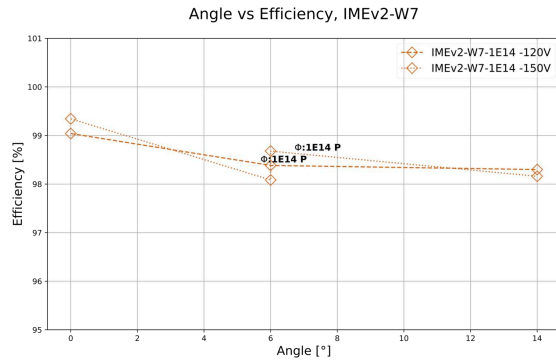
Figure 5.10: IMEv3-W12, 2×2 array, fluence $1.5 \times 10^{15} n_{eq}$. At both voltages (-450 V and -490 V) the pads exhibited very minor changes as the angles changed.



(a) Angle vs Charge

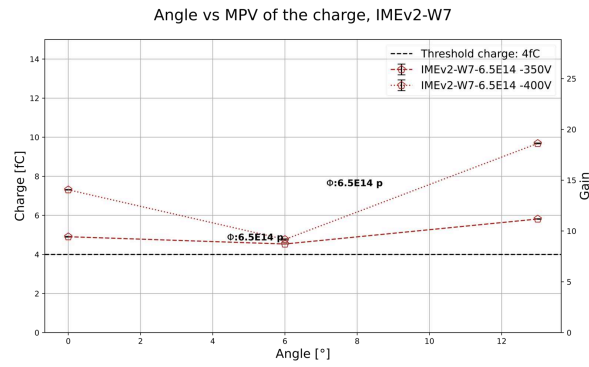


(b) Angle vs Time resolution

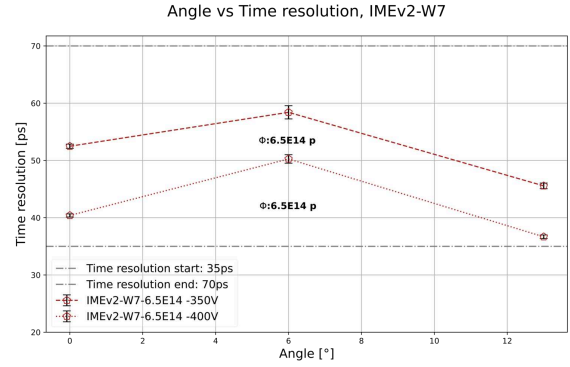


(c) Angle vs Efficiency

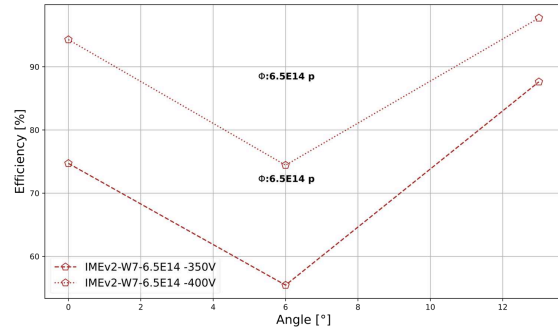
Figure 5.11: IMEv2-W7, fluence $1 \times 10^{14} n_{eq}$. Although the sensors mostly behaved as expected, two batches at the same incident angle measured different charge but comparable time resolution. The source discrepancy could not be identified. (The only potential difference was that in the batch with higher charge, the sensor was ≈ 1 mm more towards the center of the beam.)



(a) Angle vs Charge



(b) Angle vs Time resolution



(c) Angle vs Efficiency

Figure 5.12: IMEv2-W7, fluence $6.5 \times 10^{14} \text{ n}_{\text{eq}}$. This sensor exhibited one peculiarity: the data points at 6° had unexpectedly worse charge and time resolution values. Upon further investigation, in both batches the selected ROI cut out most of the sensor, greatly reducing the amount and the utility of the data. Figure 5.13 shows the selected ROI.

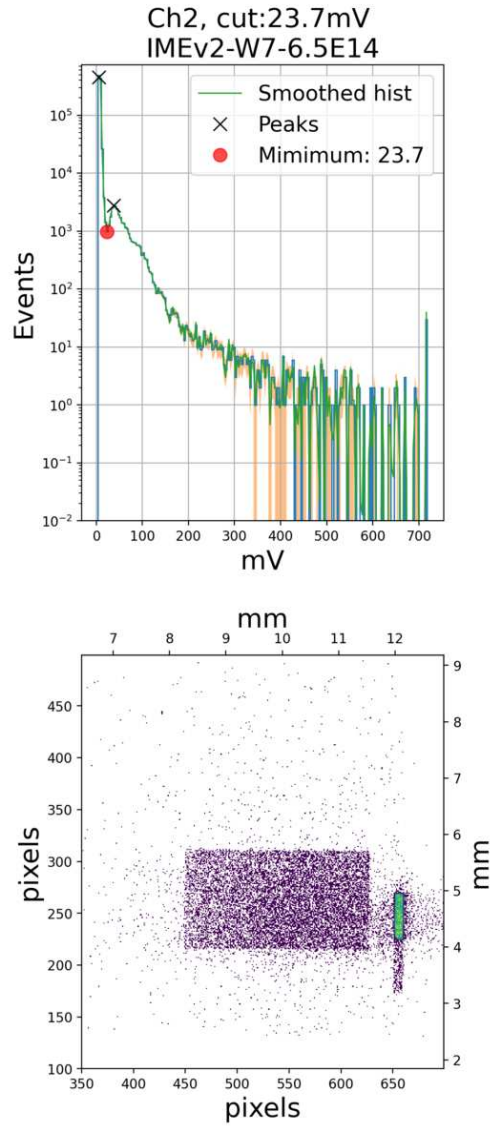
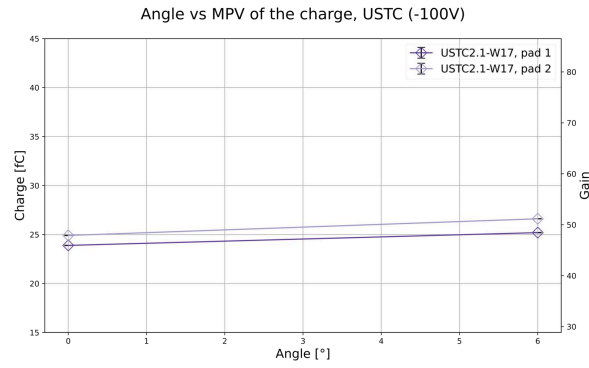
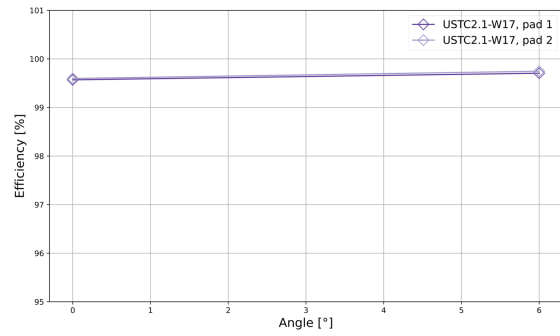


Figure 5.13: The *pulse height cut* applied to the sensor IMEv2-W7, showing how an erroneous ROI selected only a thin slice of the real pad.

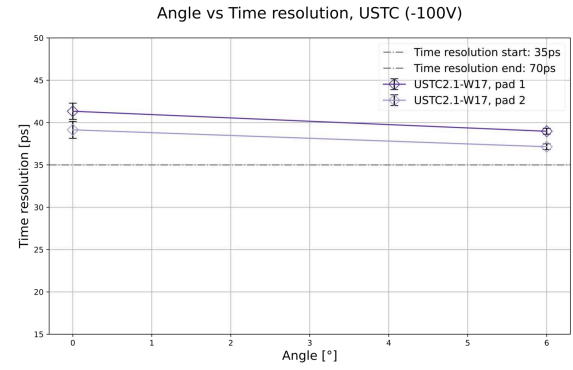
USTC



(a) Voltage vs Efficiency
Angle vs Efficiency, USTC (-100V)

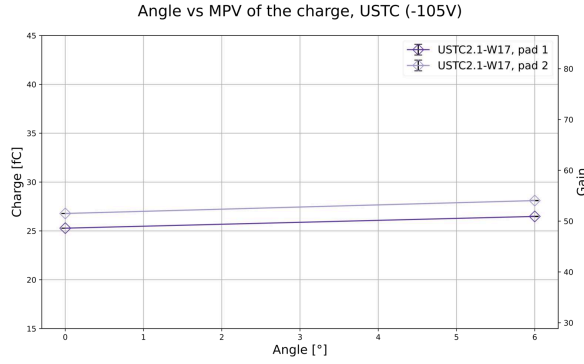


(c) Voltage vs Efficiency

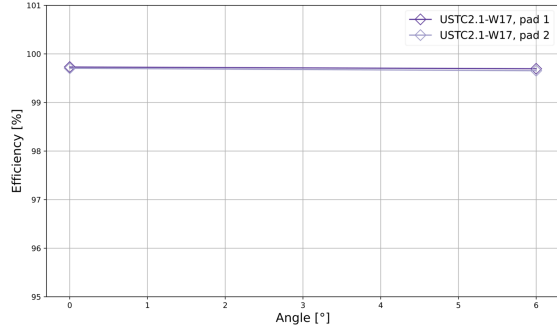


(b) Voltage vs Efficiency

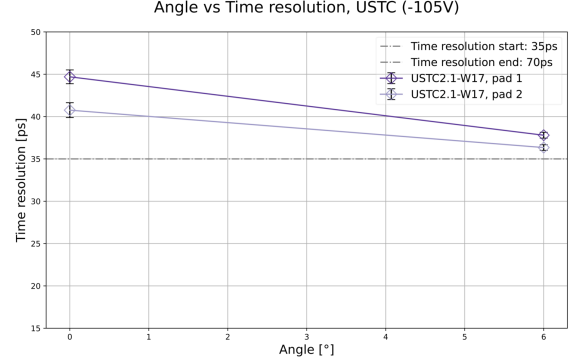
Figure 5.14: USTC, at -100 V . The sensor showed a faint dependence of its characteristics with the incident angle, and the difference between the two pads was confirmed again.



(a) Voltage vs Efficiency
Angle vs Efficiency, USTC (-105V)



(c) Voltage vs Efficiency



(b) Voltage vs Efficiency

Figure 5.15: USTC, at -105 V. Behaved very similarly to the previous measurements at -100 V.

5.2 Detailed analysis

Some other interesting effects were investigated further.

5.2.1 Noise from the edges

An anomaly in the time distribution was the asymmetry between the tails, with the left side notably diverging from a normal distribution. By picking only the events in a specific time frame, and plotting the position of the corresponding reconstructed tracks, we were able to determine that the ring surrounding the gain layer was responsible for this effect (Figure 5.16). The result was observed throughout all devices. This further justified our choice of selecting a smaller central area for most of the analysis.

5.2.2 Signal from neighbouring pads

As the momentum of the incident particles was fixed, and the distance between the MCP and the DUTs was not changed during the runs, the time of flight from MCP to DUTs should peak at one point, not two, as shown in Figure 4.6.

The nature of the second peak was investigated by checking the spatial characteristics of the associated events, i.e. selecting all the events inside this time window and checking the spatial projection of the tracks on the DUT. It was found that the main peak simply coincides with the DUT, and the second peak arises from events picked up by its neighbouring pad. Figure 5.17 is evidence of this. This second group of events was observed with a delay of the order of 1 ns. As further confirmation, this effect was only observed in DUTs which were part of LGAD arrays.

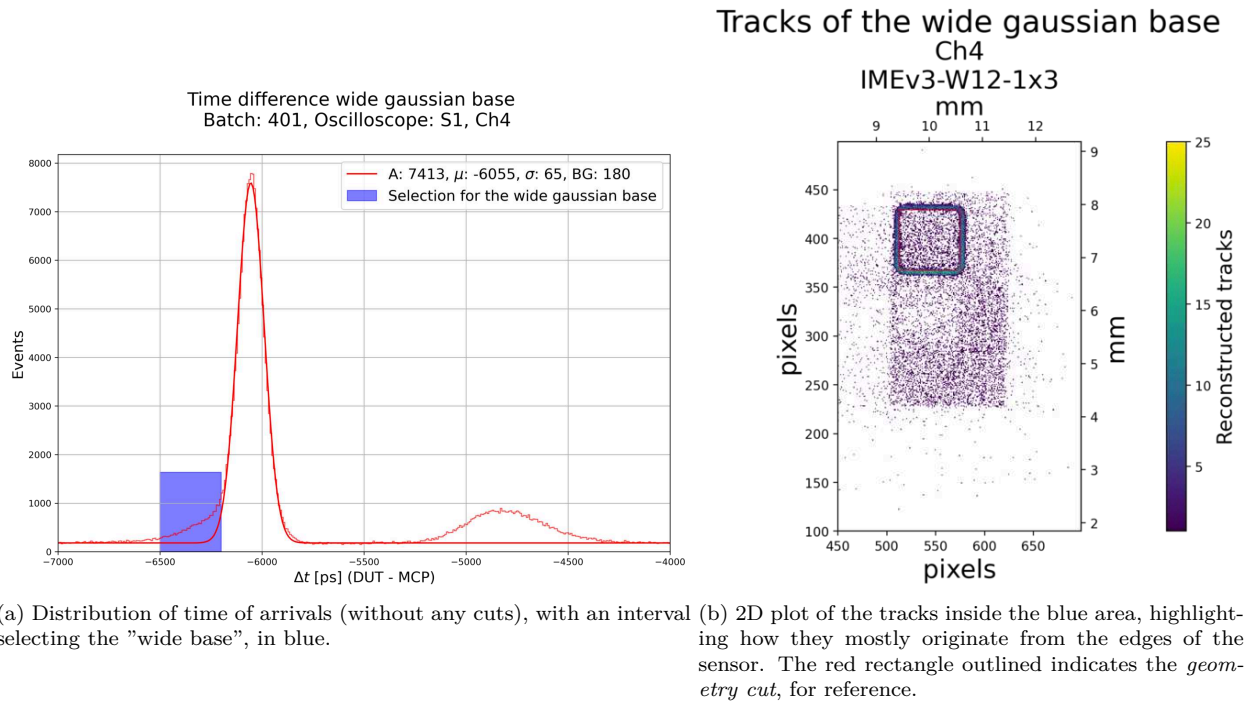
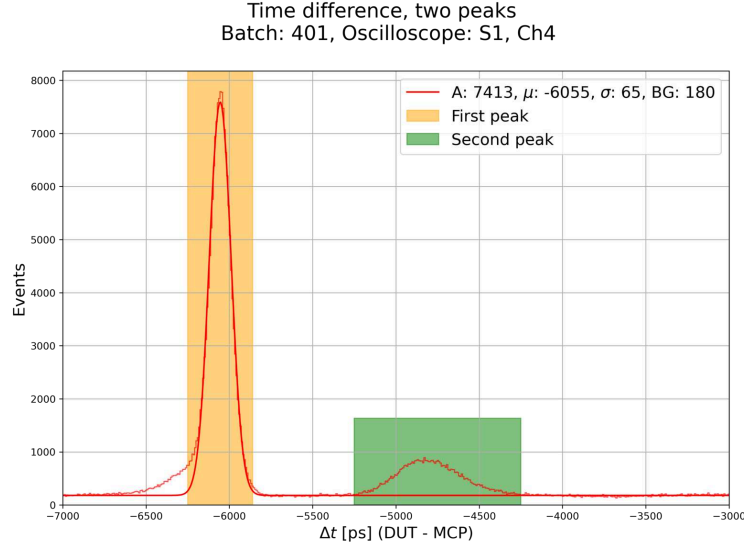
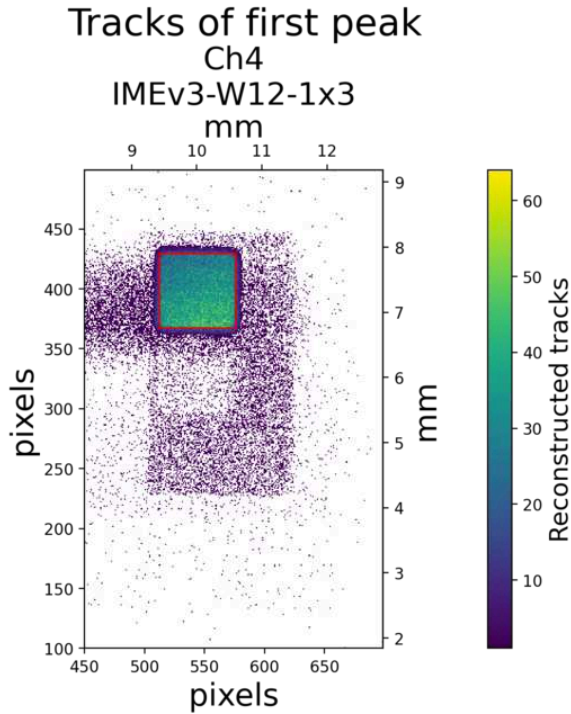


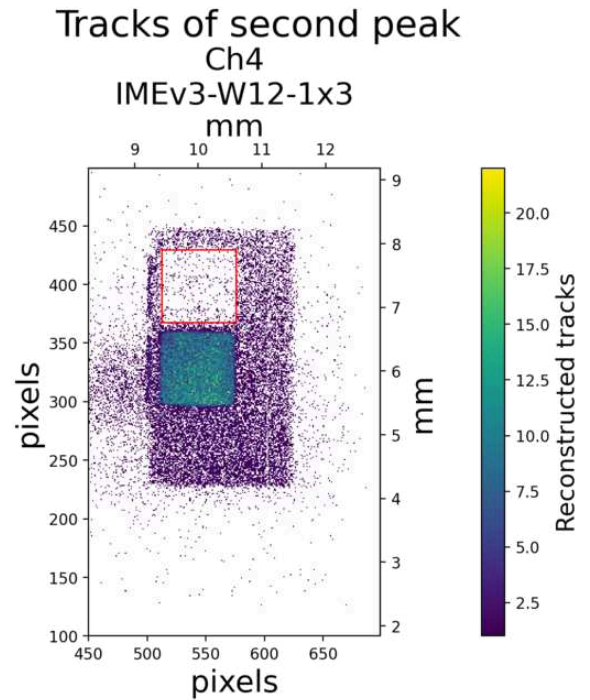
Figure 5.16: Left: time distribution of the events, with the anomalous "wide base" in blue. Right: plot of all the tracks inside the blue interval.



(a) Time distribution of the events, with the first and second peak highlighted in yellow and green, respectively.



(b) 2D plot of the tracks inside the yellow first peak, revealing the connected pad



(c) 2D plot of the tracks inside the green second peak, revealing the connected pad

Figure 5.17: Top: Time difference (DUT-MCP) distribution, with the first and second peak highlighted. Left: Tracks of events inside the yellow interval. Right: Tracks of events inside the green interval. In red is the outline of the DUT, estimated with the *geometry cut*.

5.2.3 Pulse clipping

As briefly mentioned before, there was indirect evidence that the pulses recorded by the oscilloscopes had been "cut" at their highest point. To verify this, the waveforms data was directly investigated, and a sample is shown in Figure 5.18. The apex of the pulse shows clearly a spurious plateau caused by the voltage value exceeding the range set on the oscilloscope. The consequences of this were:

- Anomaly in the pulse height distribution (Figure 4.3)
- Irregularities in the charge distribution (Figure 5.19)

Fortunately, this effect only impacted a small percentage of the data, thus it did not have a significant impact on the analysis.

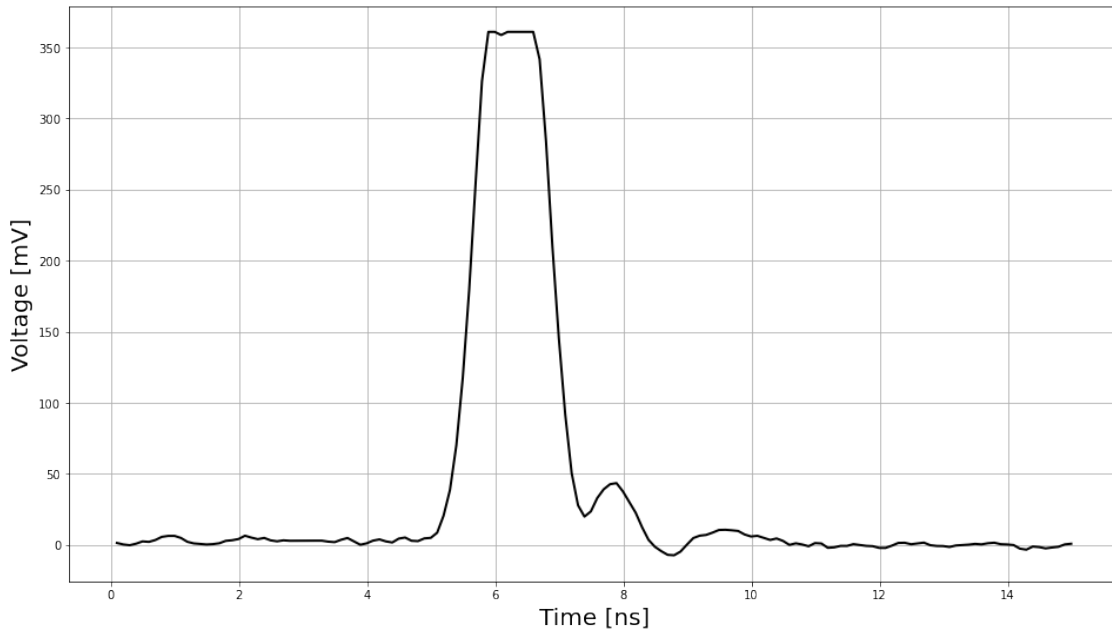


Figure 5.18: Example of a single pulse with the highest points cut out

Irregularities in the charge distribution

In a few cases the tail portion of the charge distribution deviated considerably from the expected function, left graph of Figure 5.19, as a consequence of the pulse clipping. In the right graph of Figure 5.19 the pulse height is plotted against the charge and the clipped events appear distinctly at the rightmost part of the plot. This group of events overlapped with the expected charge distribution and could not be removed without altering it. For this reason, the only solution was to adjust the range of the fit to exclude these events.

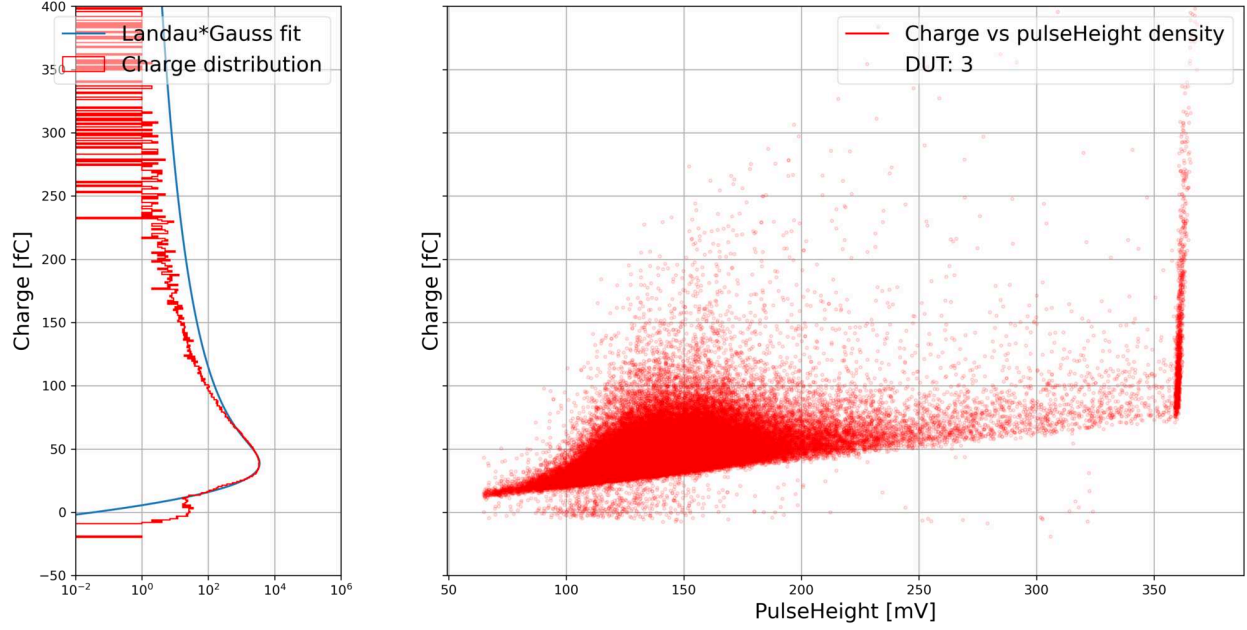
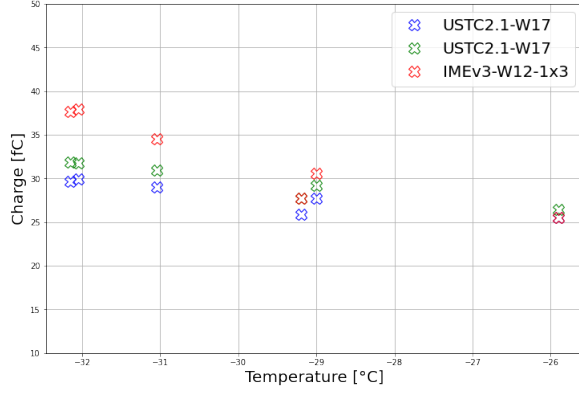


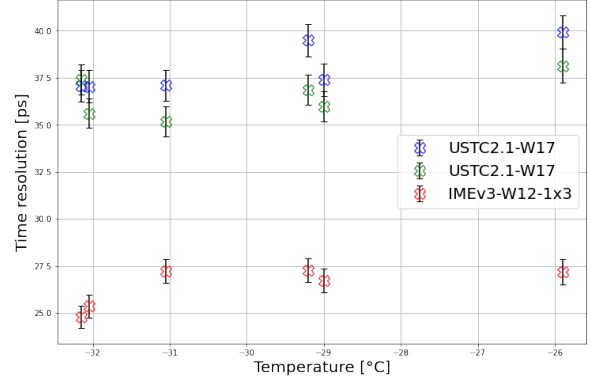
Figure 5.19: Left: distribution of collected charge with Gaussian*Landau fit, the tail shows a significant deviation between fit and data.
 Right: scatter plot of Pulse height vs Charge, revealing a dense line of events with identical pulse height, caused by the clipping.

5.2.4 Temperature fluctuations

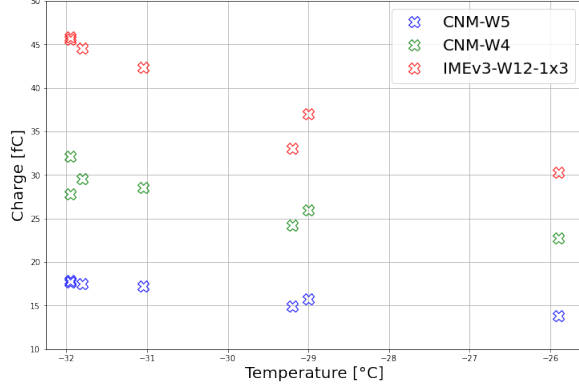
During the data taking there was a short malfunction of the cooling box containing the DUTs, which caused the temperature to rise up to $\approx -26^\circ\text{C}$. The batch was excluded from the previous analysis, but it showed that the sensors have a significant sensitivity to temperature, so it was deemed interesting enough to report. For a change in temperature of $\approx 6^\circ\text{C}$, the collected charge saw a decrease of up to $\approx 10\text{ fC}$ and time resolution saw a decrease of up to $\approx 3\text{ ps}$.



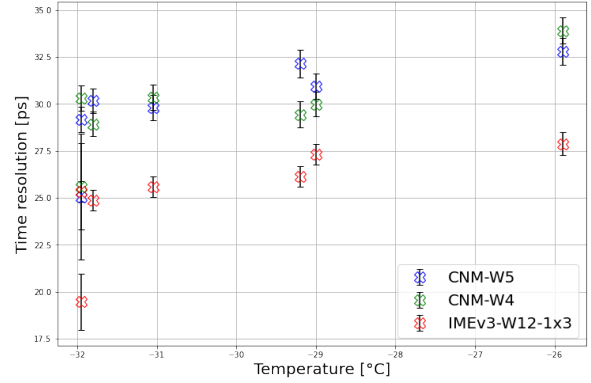
(a) Temperature vs Charge, DUTs on oscilloscope 1



(b) Temperature vs Time resolution, DUTs on oscilloscope 1



(c) Temperature vs Charge, DUTs on oscilloscope 2

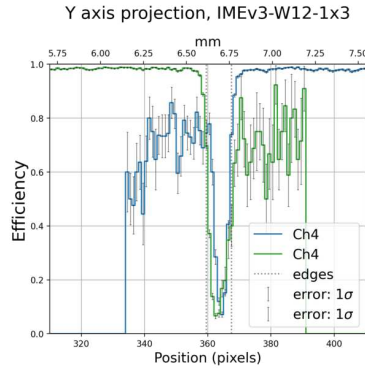


(d) Temperature vs Time resolution, DUTs on oscilloscope 2

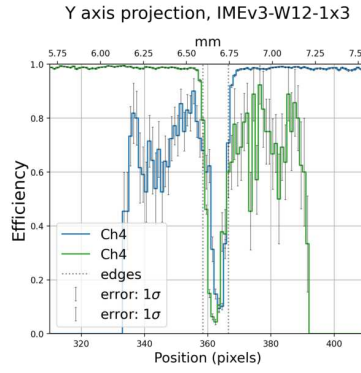
Figure 5.20: Plots of the charge and the time resolution as function of the temperature, showing a clear dependence across all sensors. NB: The sensors shown here were operated at different voltages, so this plot is not meant to be a comparison between them, rather an observation of the general trend of each of them individually.

5.2.5 Interpad study

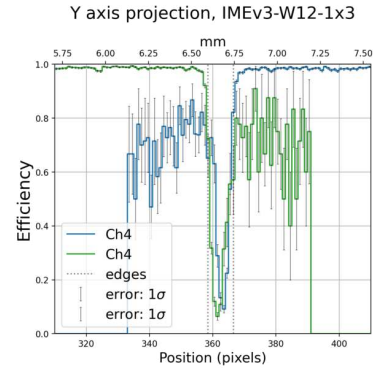
The following plots show the efficiency of the interpad region, as a function of the position. The location data was projected onto the X axis (Y axis) and, for each bin, the efficiency was calculated in a vertical (horizontal) strip. The analysis of the IMEv3-W12-2x2 was discarded due a small but significant tilt of the sensor in the XY plane (Figure A.6). This inclination meant that the projections of the positions onto one axis would not be well aligned.



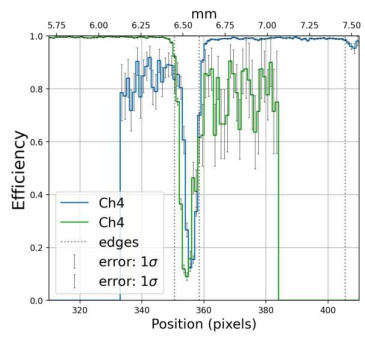
(a) Interpad efficiency, 80 V
Y axis projection, IMEv3-W12-1x3



(b) Interpad efficiency, 90 V

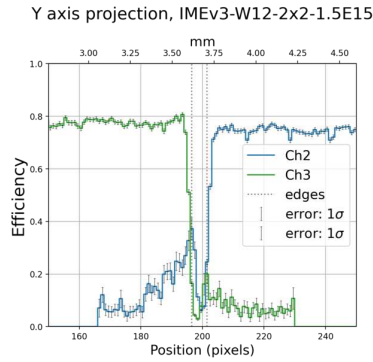


(c) Interpad efficiency, 100 V

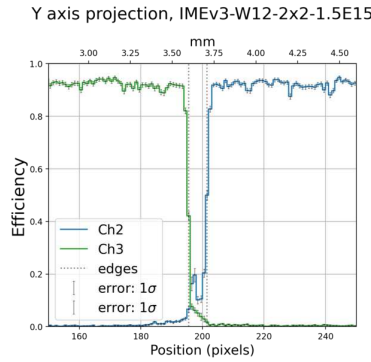


(d) Interpad efficiency, 125 V

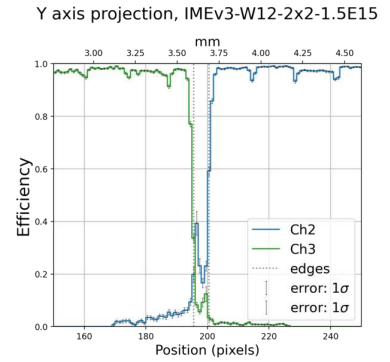
Figure 5.21: For the unirradiated IME sensor, the pads had a fairly narrow region of low efficiency, with a valley at $\approx 10\%$ efficiency, which remained mostly stable from -90 V to -125 V.



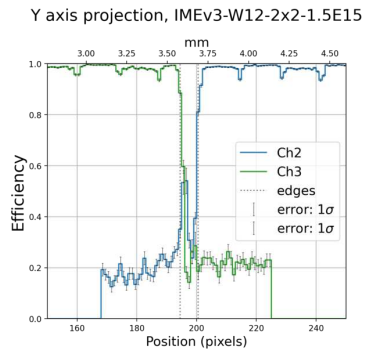
(a) Interpad efficiency, 300 V



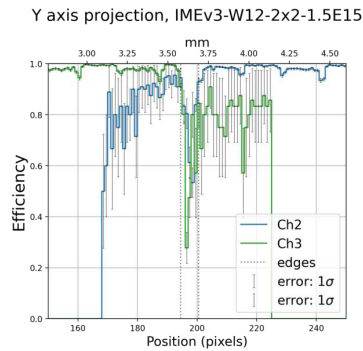
(b) Interpad efficiency, 350 V



(c) Interpad efficiency, 400 V

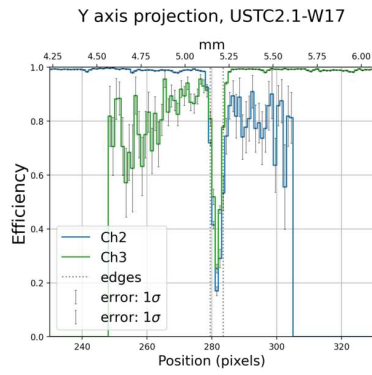


(d) Interpad efficiency, 450 V

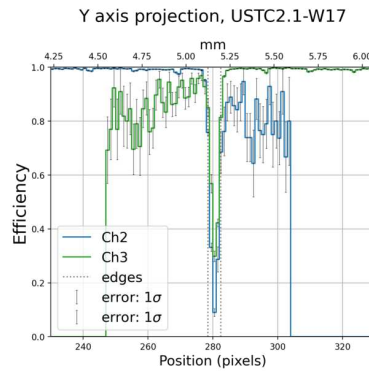


(e) Interpad efficiency, 490 V

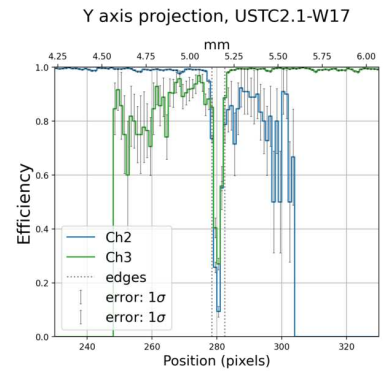
Figure 5.22: The sensor IMEv3-W12-2x2-1.5E15 showed a significant improvement of the lowest efficiency in the gap, especially above -490 V. The regularly spaced dips in efficiency, may have been caused by some physical element interfering with the beam, as this effect was observed in many sensors.



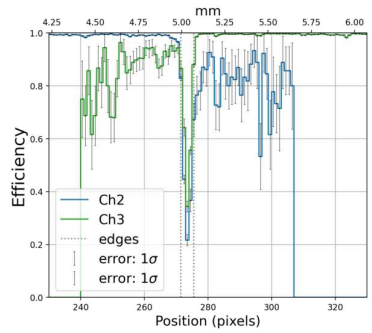
(a) Interpad efficiency, 80 V
Y axis projection, USTC2.1-W17



(b) Interpad efficiency, 90 V



(c) Interpad efficiency, 100 V



(d) Interpad efficiency, 105 V

Figure 5.23: The sensor USTC2.1-W17 displayed very little change in the interpad region, with the width and depth of the gap staying mostly constant from -80 V to -105 V.

Conclusions

This thesis had the goal of characterizing various LGADs from three manufacturers for the future HGTD, using the data from the test beam campaign of May 2023. For each sensor, the collected charge, the gain, the efficiency and the time resolution were measured, at the temperature of -30°C . The devices were divided by vendor (USTC, IHEP-IME, CNM), by version, and by radiation dose (from 0 n_{eq} to $2.5 \times 10^{15}\text{ n}_{\text{eq}}$). The analysis was at times negatively impacted by some missing data and by some erroneous selections of the FEi4 ROI.

Overall, all sensors without radiation damage (except for one: IMEv3-W12-2x2) satisfied the requirements of the HGTD:

- Collected charge $> 4\text{ fC}$.
- Per-hit efficiency $> 95\%$.
- Time resolution $< 35\text{ ps}$ to 70 ps (beginning to end of life).

All the irradiated sensors satisfied the requirements (at least for the highest tested voltages) up to a fluence of $1.5 \times 10^{15}\text{ n}_{\text{eq}}$. In particular, IMEv3-W12-2x2 irradiated at $1.5 \times 10^{15}\text{ n}_{\text{eq}}$ ($\approx 60\%$ of the total radiation dose at the end of life) had a collected charge above 8 fC and a time resolution better than 35 ps for bias voltage higher than 400 V . The results were comparable with similar tests of irradiated IHEP-IMEv2, such as in [38]. Due to missing data, the low irradiation USTC sensor could not be analyzed.

However, the two sensors at the highest fluence ($2.5 \times 10^{15}\text{ n}_{\text{eq}}$), one from IHEP and one from CNM, both fell short of the charge and efficiency targets, while still achieving a time resolution within the end of life requirement. Although, the present results show considerably worse performance than in other studies of IHEP-IMEv2, like in [38] and in [39], but compatible with the irradiated CNM studies in [36].

The sensors were also tested at an angle (0° , 6° and 14°) to the beam direction. In every case, all properties showed a slight improvement.

Additionally, for the 2x2 and 1x3 array sensors, it was shown that a significant amount of particles was detected by the neighbouring pads, generating an extra signal with a short delay of the order of 1 ns .

Although this study was conducted in the later stages of the LGAD R&D program, these results may help validate previous results and contribute to the broader understanding of LGADs behavior.

A | Appendix

A.1 On the Vavilov distribution approximation

The theoretical distribution of the energy loss of heavy particles hitting a thin target has been solved rigorously by Vavilov in [24]. The namesake distribution is a generalization of the Landau distribution, but its evaluation is more difficult, as it is expressed as an integral over some complicated functions [24, Eq.(4)]. Fortunately, it can be approximated by more straightforward distributions depending on the value of a parameter κ (Eq. (A.1)): for $\kappa \rightarrow 0$ the Vavilov distribution can be approximated by a Langau, for $\kappa \gg 1$ by a Gaussian. In this section we have replicated the calculations done in [40] applied to our specific case to ensure that the Landau approximation is valid.

A.1.1 Vavilov approximation parameter

The parameter κ is defined as:

$$\kappa = \frac{\xi}{\epsilon_{max}} \quad (\text{A.1})$$

where ξ and ϵ_{max} [40, Eq.(1)] are:

$$\xi = \frac{1}{2} K \frac{Z}{A} \frac{s}{\beta^2}; \quad s = \Delta x \cdot \rho$$

$$\epsilon_{max} = \frac{2m_e c^2 \beta^2}{1 - \beta^2} \left[1 + \frac{2m_e}{M} \frac{1}{\sqrt{1 - \beta^2}} + \left(\frac{m_e}{M} \right)^2 \right]^{-1}$$

The parameters just mentioned are the same that appear in the Bethe-Bloch mean energy loss:

$$\left\langle -\frac{dE}{dx} \right\rangle \frac{1}{\rho} = K z_0^2 \frac{1}{\beta^2} \frac{Z}{A M_u} \cdot \left[\frac{1}{2} \ln \left(\frac{2m_e c^2 \beta^2 W_{max}}{I^2 \cdot (1 - \beta^2)} \right) - \beta^2 \right]$$

And they have the following values:

$$\frac{K}{M_u} = 4\pi N_A r_e^2 m_e c^2 / M_u = 0.307075 \text{ MeV g}^{-1} \text{ cm}^2$$

$$r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2}; \quad M_u = 1 \frac{\text{g}}{\text{mol}}; \quad z_0 = 1; \quad (W_{max} \equiv \epsilon_{max})$$

For the case of **silicon**:

$$Z = 14; \quad A = 28.085; \quad \rho = 2.329085 \text{g cm}^{-3}; \quad I = 173 \text{eV}^1$$

Finally, for a Pion ($M = 139.570 \text{MeV}/c^2$) with energy $E = 120 \text{GeV}$ ($\gamma = 860.78$ and $\beta^2 = 0.99999865$) and the thin layer ($50 \mu\text{m}$) of an LGAD sensor, we obtained a value:

$$\kappa = 8.5959 \times 10^{-9} \quad (\text{A.2})$$

Well within the Landau approximation.

Reversing the calculation and fixing $\kappa = 0.01$, which is approximately the lower limit of validity, the energy of the particle should be below: 491.9MeV .

A.2 Dead pixels line

Some batches (for example Figure 4.5a) presented an empty vertical line, even before any data processing had been applied. By inspecting the data of the hits registered by each individual MIMOSA plane, it was found that one device had a dead column of pixels, which interfered with the track reconstruction. This problem turned out to be inconsequential for the analysis, except for reduced statistics in that very small region.

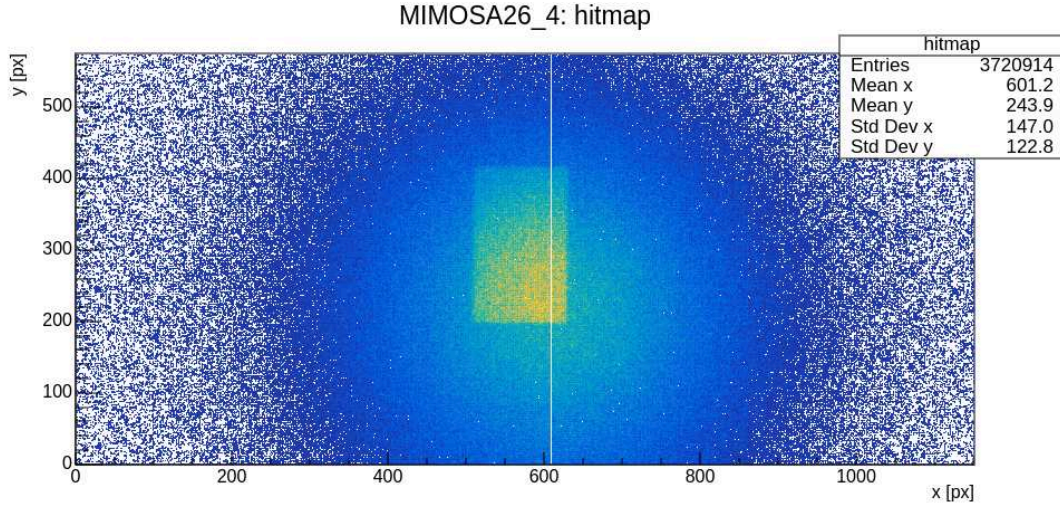


Figure A.1: The hits heatmap of the MIMOSA plane n°4, which shows clearly the dead column of pixels, the source of the "empty line" that appeared in the data.

A.3 Mislabeled sensor

One specific sensor had three channels connected to the readout board (IMEv3-W12-2x2-1.5E15, i.e. an array of 2×2 pads from IME, with fluence of $1.5 \times 10^{15} \text{n}_{\text{eq}}$), but, when plotting the outline of each pad, one of them did not match the expected configuration. Additionally, in the logbook of the test beam of May 2023, there was some contradicting information regarding the connected channels on the readout board. We suspect that the inconsistent sensor could actually be one of the pads of IMEv3-W12-**1x3**-1.5E15 but, for an unfortunate coincidence, that sensor had no useful batches. Therefore, it was not possible to confirm that

¹Mean excitation Energy

the two channels were simply swapped, and Channel4 of the IMEv3-W12-2x2-1.5E15 was excluded from the analysis.

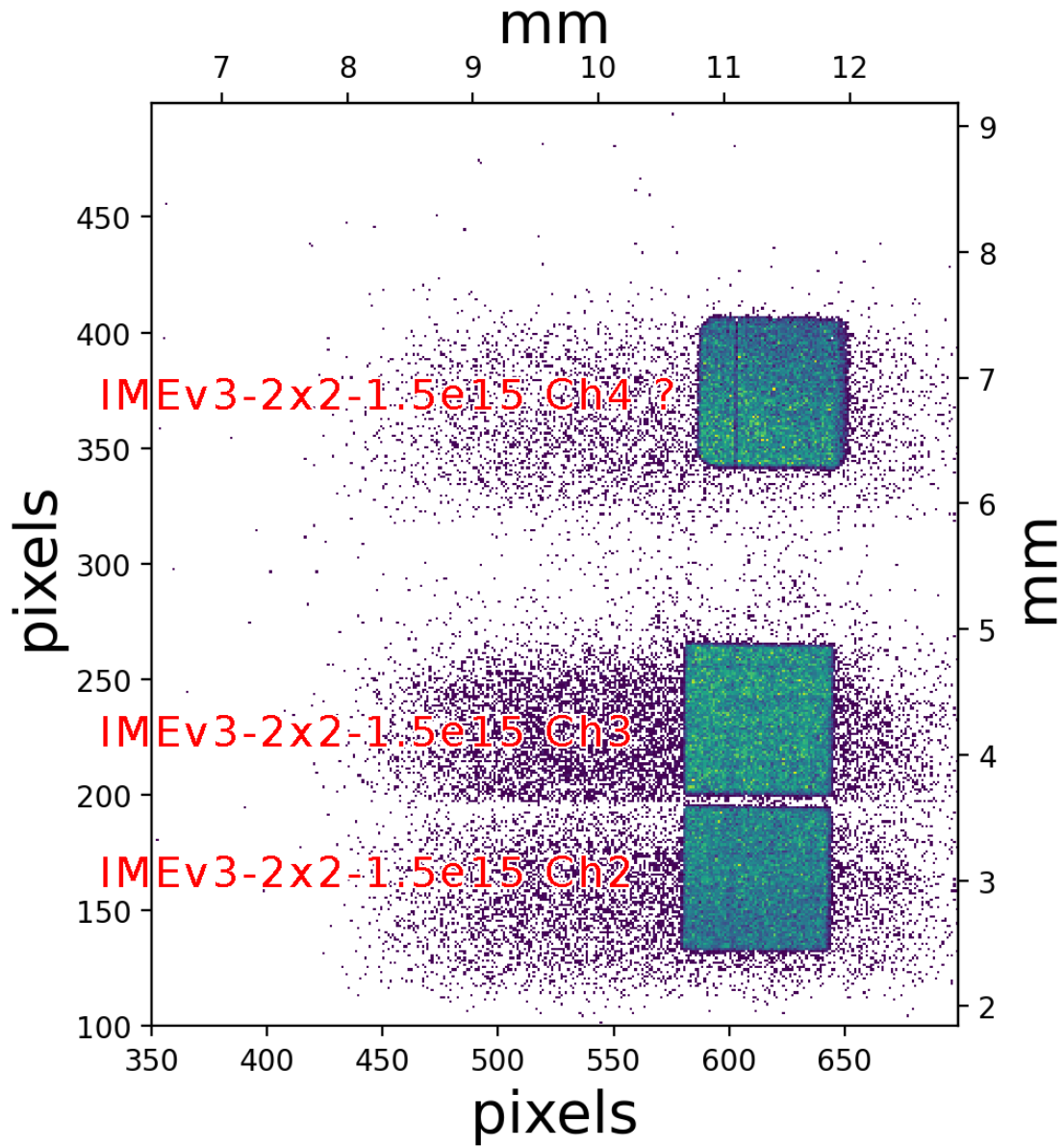


Figure A.2: This plot combines the 2D plot of the reconstructed tracks (after a *pulse height cut*) of the three channels of IMEv3-W12-2x2-1.5E15. It is very obvious that Channel4 does not match the expected shape of a 2x2 grid, and it is very likely wrongly labeled.

A.4 Supplementary Figures

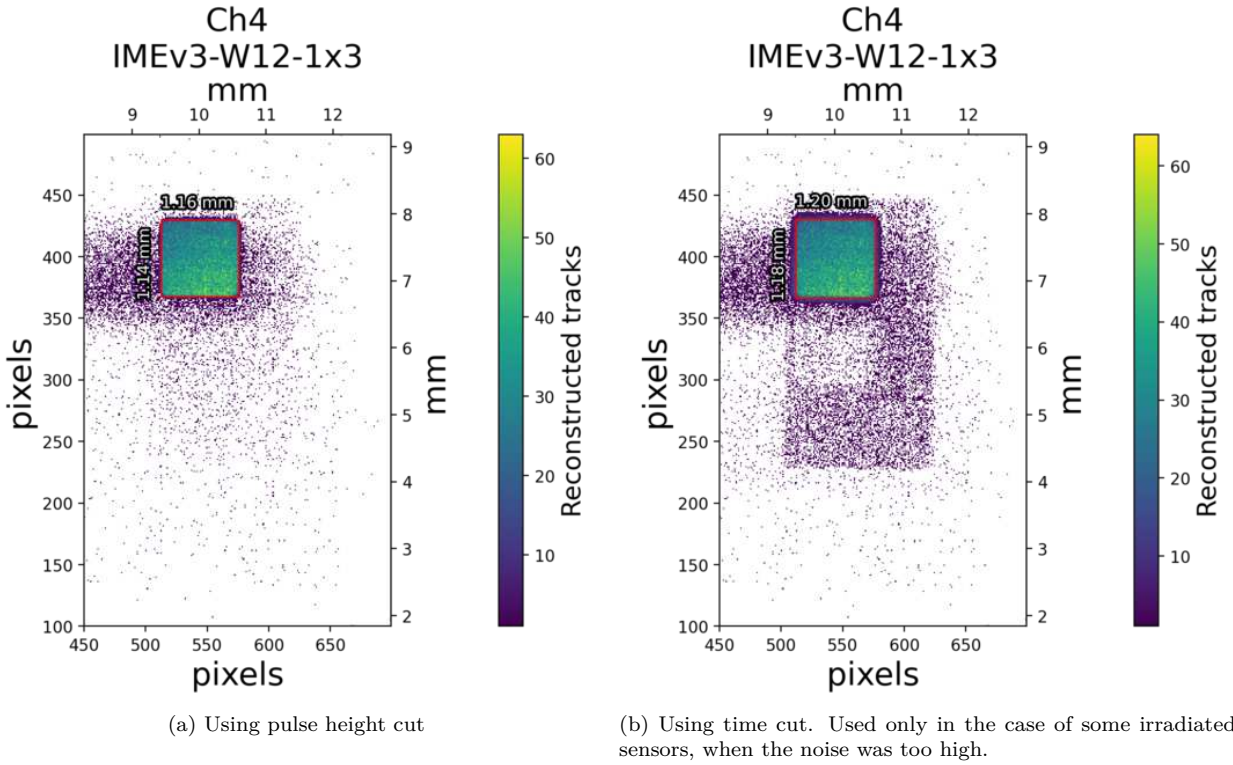


Figure A.3: Example of the *geometry cut* (red rectangles) obtained by applying a *pulse height cut* (left) and a *time cut* (left) to show that they produce comparable results. In practice, since all results were calculated inside the smaller $0.5 \times 0.5 \text{ mm}^2$ area, there were no real differences.

CFD_comparison, batch: 401 central area cuts S1 dut: 3

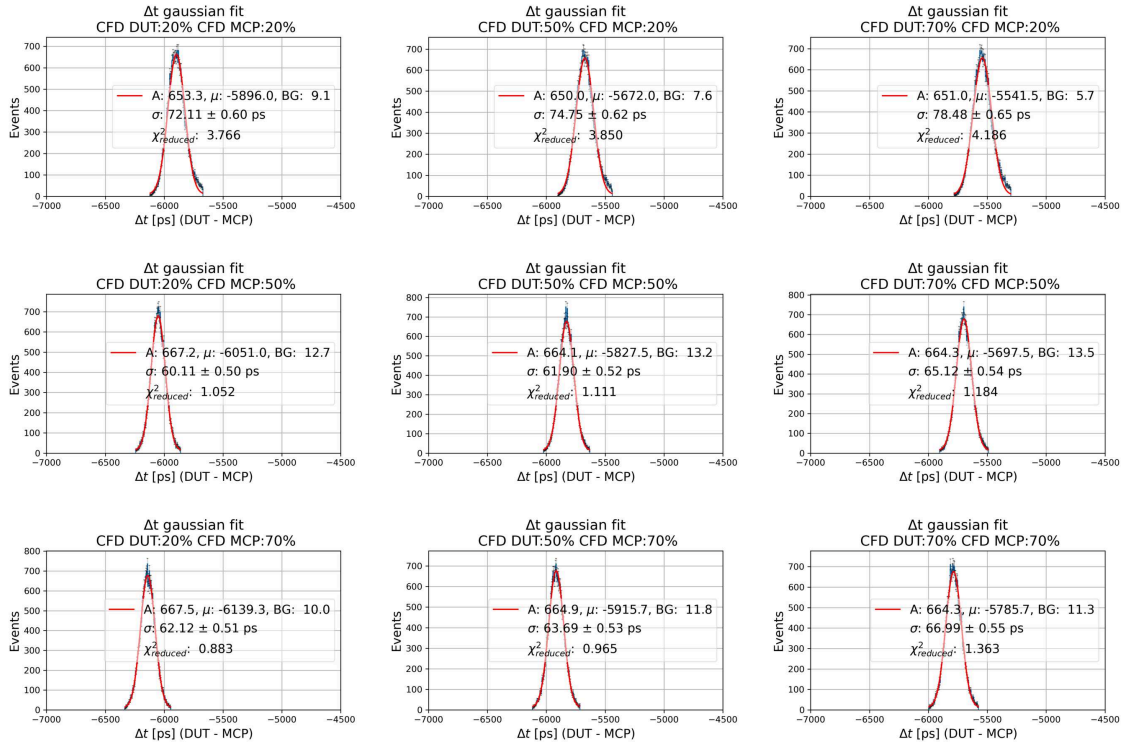


Figure A.4: Comparison of the fit for the time resolution of an unirradiated sensor (IMEv3-W12-1x3), for three different values of CFD: 20%, 50% and 70%

CFD_comparison, batch: 605 central area cuts S1 dut: 1

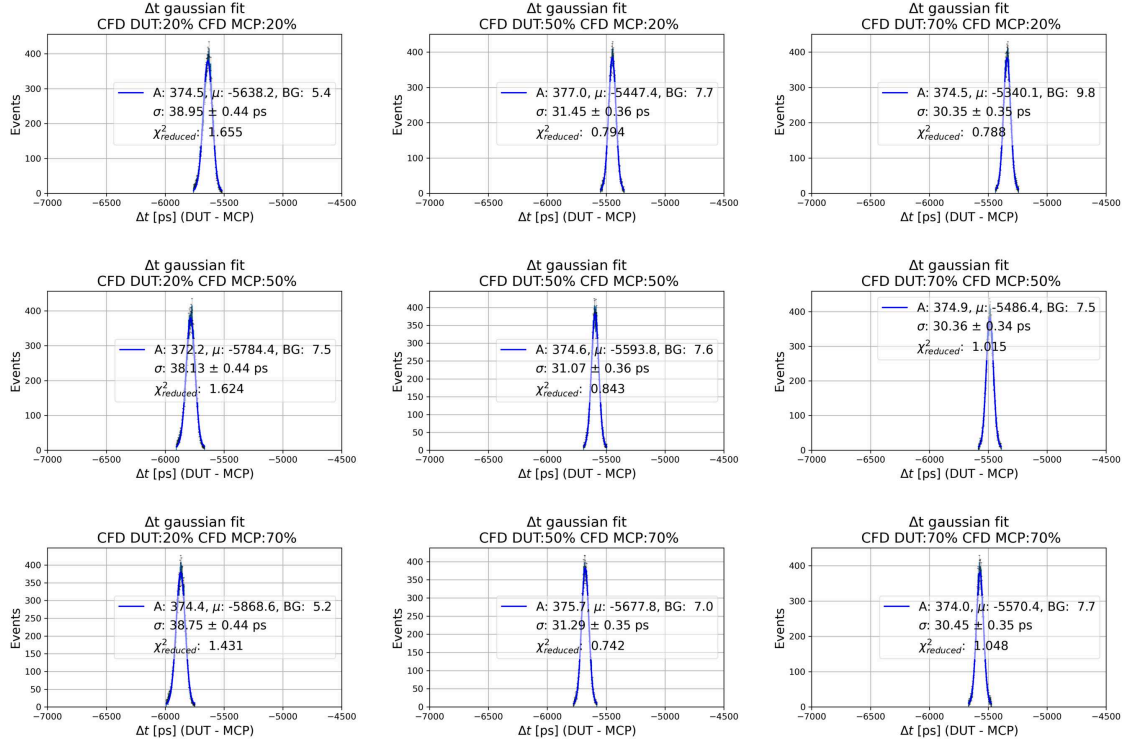


Figure A.5: Comparison of the fit for the time resolution of an irradiated sensor (IMEv3-W12-2x2-1.5E15), for three different values of CFD: 20%, 50% and 70%

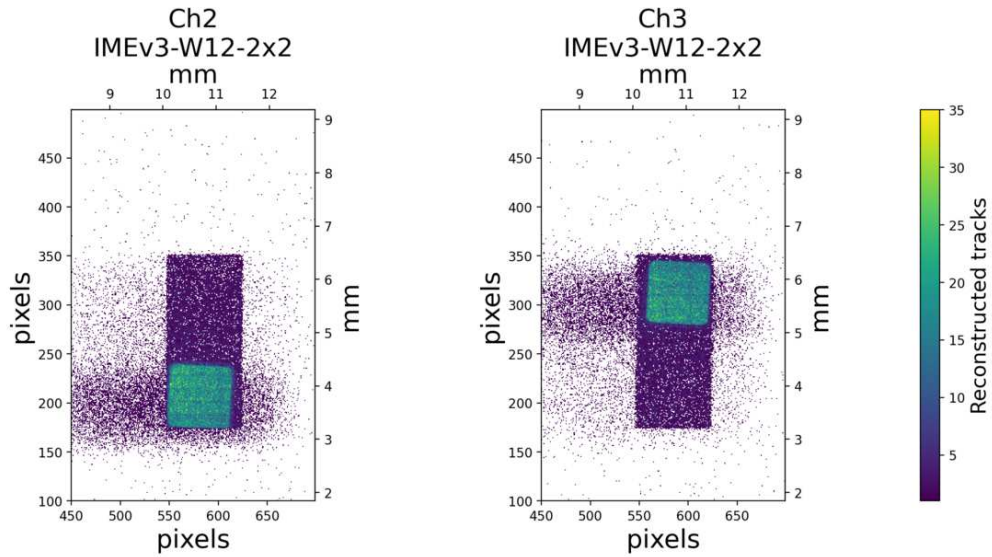


Figure A.6: An example of two sensors slightly tilted.

A.5 Tables of results

Here are reported the tables containing all the results, divided by sensor. The column "Batch" indicates the batch number, the oscilloscope ('S1' or 'S2') and the channel number; multiple entries of a single batch number refer to multiple connected pads. The batch numbered 5031 was originally numbered 503, but it was divided into two batches because there was a significant shift in the position of the DUTs. 5032 was later removed due to the redundancy and higher noise.

Table A.1: Summary of the results for the sensor: CNM-W4.

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
199	S2	Ch3	2600	-31.2	0	-80	10.39	0.03	51.73	1.84	0.990
301	S2	Ch3	2500	21.8	0	-150	13.32	0.01	38.01	0.96	0.995
401	S2	Ch3	2500	-30.6	0	-80	10.33	0.01	43.17	0.82	0.993
402	S2	Ch3	2500	-29.9	0	-100	14.27	0.01	30.68	1.09	0.994
407	S2	Ch3	2500	-30.0	0	-120	24.98	0.02	26.78	1.21	0.996
408	S2	Ch3	2500	21.4	0	-160	16.20	0.02	36.41	1.18	0.994
409	S2	Ch3	2500	21.7	0	-175	23.61	0.03	27.76	1.37	0.995
410	S2	Ch3	2600	21.8	0	-175	23.66	0.04	30.55	0.70	0.995
411	S2	Ch3	2500	-29.6	13	-120	28.94	0.03	25.71	1.44	0.997
413	S2	Ch3	2800	-30.2	6	-120	25.60	0.03	30.04	0.27	0.997
414	S2	Ch3	2800	-31.9	6	-100	15.03	0.01	41.60	0.31	0.997

Table A.2: Summary of the results for the sensor: CNM-W5.

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
199	S2	Ch2	2600	-31.2	0	-80	9.70	0.06	50.32	3.95	0.992
301	S2	Ch2	2500	21.8	0	-180	19.11	0.01	29.02	1.17	0.994
401	S2	Ch2	2500	-30.6	0	-80	9.75	0.01	44.43	0.80	0.992
402	S2	Ch2	2500	-29.9	0	-100	12.60	0.01	33.53	1.02	0.993
403	S2	Ch2	2500	-31.2	0	-120	18.39	0.02	25.21	1.36	0.993
407	S2	Ch2	2500	-30.0	0	-135	22.99	0.01	27.78	1.16	0.995
408	S2	Ch2	2500	21.4	0	-185	13.63	0.02	33.27	1.24	0.993
409	S2	Ch2	2500	21.7	0	-195	17.31	0.02	27.11	1.39	0.993
410	S2	Ch2	2600	21.8	0	-195	27.64	0.04	29.74	0.66	0.997
411	S2	Ch2	2500	-29.6	13	-130	28.67	0.03	26.11	1.39	0.995
413	S2	Ch2	2800	-30.2	6	-130	15.50	0.02	30.24	0.28	0.995
414	S2	Ch2	2800	-31.9	6	-120	11.85	0.01	34.14	0.28	0.996

Table A.3: Summary of the results for the sensor: CNM-W5-1.5E15.

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
501	S2	Ch3	2800	-26.7	0	-400	2.37	0.01	56.80	0.60	0.306
502	S2	Ch3	2800	-26.6	0	-450	3.07	0.01	53.18	0.44	0.476
504	S2	Ch3	2800	-33.2	0	-550	6.46	0.01	37.42	0.28	0.948
505	S2	Ch3	2800	-33.1	0	-600	10.54	0.01	31.17	0.27	0.988
1201	S1	Ch3	2800	0.0	0	-620	9.11	0.01	30.61	0.55	0.947
1202	S1	Ch3	2800	0.0	0	-640	11.14	0.03	28.84	0.52	0.963
5031	S2	Ch3	2800	-29.8	0	-500	4.26	0.01	45.74	0.44	0.727

Table A.4: Summary of the results for the sensor: CNM-W3-2.5E15.

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
601	S2	Ch3	2800	-31.0	0	-400	1.57	0.01	31.02	1.98	0.048
602	S2	Ch3	2800	-30.8	0	-450	1.53	0.00	39.66	1.91	0.049
603	S2	Ch3	2800	-31.0	0	-500	1.90	0.00	56.00	2.09	0.057
604	S2	Ch3	2800	-31.1	0	-550	1.43	0.01	63.40	2.04	0.080
605	S2	Ch3	2800	-31.0	0	-600	2.65	0.00	62.29	1.62	0.253
1201	S1	Ch2	2800	0.0	0	-620	2.12	0.01	56.12	1.84	0.235
1202	S1	Ch2	2800	0.0	0	-640	2.51	0.04	44.78	5.93	0.358

Table A.5: Summary of the results for the sensor: USTC2.1-W17.

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
301	S1	Ch2	2500	21.8	0	-120	11.00	0.01	61.70	0.79	0.994
		Ch3	2500	21.8	0	-120	11.51	0.01	55.00	0.85	0.996
401	S1	Ch2	2500	-30.6	0	-80	17.02	0.01	52.38	0.74	0.995
		Ch3	2500	-30.6	0	-80	17.72	0.01	46.48	0.80	0.996
402	S1	Ch2	2500	-29.9	0	-90	19.44	0.01	47.40	0.81	0.996
		Ch3	2500	-29.9	0	-90	20.19	0.01	43.01	0.88	0.996
403	S1	Ch2	2500	-31.2	0	-100	23.89	0.02	41.34	0.96	0.996
		Ch3	2500	-31.2	0	-100	24.91	0.02	39.14	1.00	0.996
407	S1	Ch2	2500	-30.0	0	-105	25.28	0.01	44.70	0.81	0.997
		Ch3	2500	-30.0	0	-105	26.78	0.02	40.76	0.88	0.997
408	S1	Ch2	2500	21.4	0	-150	16.17	0.02	53.10	1.06	0.996
		Ch3	2500	21.4	0	-150	16.92	0.02	49.03	1.18	0.996
409	S1	Ch2	2500	21.7	0	-165	20.64	0.02	46.65	1.08	0.996
		Ch3	2500	21.7	0	-165	21.89	0.02	44.83	1.14	0.997
410	S1	Ch2	2600	21.8	0	-165	20.65	0.03	41.31	0.77	0.995
		Ch3	2600	21.8	0	-165	21.90	0.03	40.87	0.86	0.996
413	S1	Ch2	2800	-30.2	6	-105	26.48	0.03	37.80	0.34	0.997
		Ch3	2800	-30.2	6	-105	28.10	0.03	36.34	0.33	0.997
414	S1	Ch2	2800	-31.9	6	-100	25.20	0.02	38.97	0.35	0.997
		Ch3	2800	-31.9	6	-100	26.61	0.02	37.15	0.34	0.997

Table A.6: Summary of the results for the sensor: IMEv3-W12-2x2.

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
199	S1	Ch2	2600	-31.2	0	-80	13.75	0.09	77.22	5.17	0.994
		Ch3	2600	-31.2	0	-80	14.00	0.09	69.80	2.70	0.983
201	S1	Ch2	2500	-31.8	0	-80	13.97	0.01	50.30	0.91	0.995
		Ch3	2500	-31.8	0	-80	14.23	0.02	51.82	0.94	0.994
202	S1	Ch2	2500	-32.5	0	-90	16.38	0.01	46.06	0.91	0.995
		Ch3	2500	-32.5	0	-90	16.48	0.02	49.11	0.92	0.995
203	S1	Ch2	2500	-32.6	0	-100	19.64	0.02	42.70	1.06	0.989
		Ch3	2500	-32.6	0	-100	19.45	0.03	45.80	1.09	0.990
204	S1	Ch2	2500	-30.3	0	-125	49.00	0.08	42.00	0.96	0.999
205	S1	Ch2	2500	-28.2	13	-125	46.13	0.05	21.43	1.55	0.998
206	S1	Ch2	2500	-27.9	13	-100	19.94	0.02	36.70	1.18	0.996
		Ch3	2500	-27.9	13	-100	20.30	0.03	35.99	1.20	0.995

Table A.7: Summary of the results for the sensor: IMEv3-W12-1x3

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
199	S1	Ch4	2600	-31.2	0	-80	11.53	0.04	56.62	1.98	0.995
	S2	Ch4	2600	-31.2	0	-80	13.65	0.04	49.81	1.82	0.987
301	S2	Ch4	2500	21.8	0	-100	7.65	0.01	59.25	0.87	0.986
401	S1	Ch4	2500	-30.6	0	-80	11.39	0.01	47.72	0.88	0.994
	S2	Ch4	2500	-30.6	0	-80	13.60	0.01	41.02	0.87	0.995
402	S1	Ch4	2500	-29.9	0	-90	12.93	0.01	43.20	0.98	0.994
	S2	Ch4	2500	-29.9	0	-90	15.44	0.01	36.92	0.97	0.995
403	S1	Ch4	2500	-31.2	0	-100	15.47	0.02	39.13	1.14	0.994
	S2	Ch4	2500	-31.2	0	-100	18.55	0.02	31.76	1.17	0.995
407	S1	Ch4	2500	-30.0	0	-125	29.73	0.03	32.29	1.09	0.997
	S2	Ch4	2500	-30.0	0	-125	35.66	0.03	21.31	1.50	0.997
408	S1	Ch4	2500	21.4	0	-160	13.64	0.03	42.42	1.34	0.995
	S2	Ch4	2500	21.4	0	-160	16.50	0.02	32.42	1.33	0.995
409	S1	Ch4	2500	21.7	0	-160	13.58	0.03	40.82	1.35	0.990
	S2	Ch4	2500	21.7	0	-160	16.39	0.02	32.17	1.32	0.995
410	S1	Ch4	2600	21.8	0	-160	13.56	0.04	39.80	1.04	0.990
	S2	Ch4	2600	21.8	0	-160	16.46	0.03	34.05	0.82	0.994
411	S1	Ch4	2500	-29.6	13	-125	34.25	0.07	33.41	1.33	0.997
	S2	Ch4	2500	-29.6	13	-125	41.26	0.07	22.30	1.63	0.997
413	S1	Ch4	2800	-30.2	6	-125	29.16	0.04	26.17	0.30	0.998
	S2	Ch4	2800	-30.2	6	-125	36.46	0.05	25.58	0.27	0.997
414	S1	Ch4	2800	-31.9	6	-100	15.75	0.01	34.28	0.34	0.996
	S2	Ch4	2800	-31.9	6	-100	18.96	0.02	33.66	0.27	0.995

Table A.8: Summary of the results for the sensor: IMEv3-W12-2x2-1.5E15

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
601	S1	Ch2	2800	-31.0	0	-300	4.46	0.01	47.37	0.39	0.764
		Ch3	2800	-31.0	0	-300	4.69	0.01	48.30	0.43	0.792
602	S1	Ch2	2800	-30.8	0	-350	5.96	0.02	39.56	0.49	0.932
		Ch3	2800	-30.8	0	-350	7.84	0.00	41.96	0.56	0.940
603	S1	Ch2	2800	-31.0	0	-400	8.68	0.01	34.95	0.34	0.984
		Ch3	2800	-31.0	0	-400	9.10	0.01	36.03	0.38	0.984
604	S1	Ch2	2800	-31.1	0	-450	12.24	0.01	30.84	0.30	0.987
		Ch3	2800	-31.1	0	-450	12.75	0.02	31.60	0.32	0.989
605	S1	Ch2	2800	-31.0	0	-490	16.37	0.03	30.24	0.38	0.985
		Ch3	2800	-31.0	0	-490	17.09	0.03	30.71	0.43	0.985
701	S1	Ch2	2800	-31.3	14	-490	19.57	0.02	29.02	0.24	0.988
		Ch3	2800	-31.3	14	-490	20.35	0.02	30.29	0.26	0.987
702	S1	Ch2	2800	-31.1	14	-450	14.06	0.02	30.22	0.31	0.987
		Ch3	2800	-31.1	14	-450	14.70	0.02	31.96	0.34	0.988
1001	S1	Ch2	2800	-31.5	6	-490	18.63	0.02	29.43	0.30	0.988
		Ch3	2800	-31.5	6	-490	19.29	0.03	30.22	0.32	0.989
1002	S1	Ch2	2800	-31.6	6	-450	13.78	0.02	31.16	0.30	0.987
		Ch3	2800	-31.6	6	-450	14.40	0.02	31.21	0.33	0.988

Table A.9: Summary of the results for the sensor: IMEv2-W7-1E14

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
501	S2	Ch2	2800	-26.7	0	-80	5.11	0.01	73.01	0.39	0.846
502	S2	Ch2	2800	-26.6	0	-100	6.39	0.01	55.28	0.28	0.941
504	S2	Ch2	2800	-33.2	0	-150	13.91	0.01	30.18	0.24	0.992
801	S2	Ch3	2800	-31.2	14	-150	13.71	0.02	32.03	0.36	0.981
802	S2	Ch3	2800	-31.2	14	-120	8.58	0.01	43.36	0.43	0.978
1001	S2	Ch3	2800	-31.5	6	-150	11.43	0.02	30.83	0.34	0.980
1002	S2	Ch3	2800	-31.6	6	-120	9.14	0.02	42.64	0.53	0.981
1102	S2	Ch3	2800	-31.1	6	-150	14.33	0.02	31.91	0.30	0.986
5031	S2	Ch2	2800	-29.8	0	-120	8.07	0.01	44.77	0.34	0.978

Table A.10: Summary of the results for the sensor: IMEv2-W7-6.5E14

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
502	S1	Ch2	2800	-26.6	0	-250	2.65	0.01	73.86	0.76	0.250
504	S1	Ch2	2800	-33.2	0	-350	4.91	0.02	52.48	0.37	0.747
505	S1	Ch2	2800	-33.1	0	-400	7.31	0.02	40.40	0.27	0.943
901	S1	Ch2	2800	-31.3	13	-400	9.68	0.02	36.67	0.32	0.977
902	S1	Ch2	2800	-31.3	13	-350	5.81	0.02	45.56	0.49	0.876
1101	S1	Ch2	2800	-31.0	6	-400	4.77	0.03	50.27	0.74	0.744
1102	S1	Ch2	2800	-31.1	6	-350	4.54	0.00	58.41	1.15	0.554
5031	S1	Ch2	2800	-29.8	0	-300	3.65	0.00	65.44	0.85	0.425

Table A.11: Summary of the results for the sensor: IMEv3-W16-8E14

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
501	S1	Ch3	2800	-26.7	0	-200	5.23	0.00	54.49	0.29	0.798
502	S1	Ch3	2800	-26.6	0	-250	7.39	0.01	41.67	0.22	0.973
504	S1	Ch3	2800	-33.2	0	-350	16.70	0.02	30.04	0.24	0.990
505	S1	Ch3	2800	-33.1	0	-370	17.86	0.03	33.27	0.25	0.951
801	S2	Ch2	2800	-31.2	14	-360	13.90	0.03	32.05	0.37	0.957
802	S2	Ch2	2800	-31.2	14	-340	12.73	0.02	30.07	0.34	0.973
1001	S2	Ch2	2800	-31.5	6	-360	18.24	0.03	32.29	0.26	0.966
1002	S2	Ch2	2800	-31.6	6	-340	16.69	0.02	29.88	0.29	0.979
1101	S2	Ch2	2800	-31.0	6	-360	18.23	0.02	32.02	0.24	0.965
1102	S2	Ch2	2800	-31.1	6	-340	16.71	0.02	30.22	0.27	0.976
5031	S1	Ch3	2800	-29.8	0	-300	11.82	0.01	32.75	0.29	0.992

Table A.12: IMEv3-W16-2.5E15

Batch			MCP [V]	Temp. [°C]	Angle [°]	Volt. [V]	Charge [fC]	$\pm\sigma$ [fC]	Time res. [ps]	$\pm\sigma$ [ps]	Eff.
501	S1	Ch4	2800	-26.7	0	-400	2.21	0.01	57.90	0.68	0.211
502	S1	Ch4	2800	-26.6	0	-450	2.50	0.00	58.32	0.57	0.305
5031	S1	Ch4	2800	-29.8	0	-500	3.08	0.01	53.51	0.70	0.445

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